

Black Holes and Phase Saturation

Phase Theory — Paper 13

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Abstract

We reformulate black holes within Phase Theory as regions of **phase saturation**, where admissible phase curvature and distinguishability reach maximal density under global consistency constraints. Black holes are not spacetime singularities, information-destroying objects, or field-theoretic vacua, but stable phase configurations characterized by saturated admissibility. Event horizons are shown to be emergent **phase distinguishability boundaries**, not physical surfaces. Information is neither destroyed nor duplicated; it becomes externally inaccessible due to phase indistinguishability rather than loss. Hawking radiation, entropy scaling, and black hole thermodynamics emerge as effective consequences of phase saturation without invoking quantum fields on curved spacetime. We derive the **phase-theoretic no-singularity theorem**,

establish a structural resolution of the information paradox, provide phase-theoretic analogues of the Kerr and Reissner-Nordström families, and enumerate eight classes of falsifiable observational predictions with associated experimental protocols. The paper situates these results within the broader Phase Theory framework established in Papers 1-12 and compares them systematically with general relativity, string-theoretic approaches, loop quantum gravity, and holographic frameworks.

Contents

- 1.** The Black Hole Problem Reframed
- 2.** Phase Theory Foundations — A Minimal Review
- 3.** Phase Saturation — Core Concept and Formal Definition
- 4.** Black Holes as Saturated Phase Configurations
- 5.** The Phase Admissibility Tensor and Saturation Conditions
- 6.** Event Horizons as Phase Distinguishability Boundaries
- 7.** The Phase Horizon Metric — Emergent Geometry Near Saturation
- 8.** Entropy of Black Holes — Phase-Theoretic Derivation
- 9.** Area Scaling Laws and the Holographic Principle
- 10.** Hawking Radiation Without Quantum Fields
- 11.** Phase Relaxation Spectrum and Effective Temperature
- 12.** Black Hole Evaporation as Phase Density Reduction
- 13.** Information Retention and the Non-Loss Theorem
- 14.** The Page Curve from Phase Equilibration

15.	Why Radiation Appears Thermal — Observational Coarse-Graining
16.	Firewalls and Complementarity — Resolution via Global Phase Topology
17.	Interior Structure — Phase Topology Beyond the Horizon
18.	The No-Singularity Theorem
19.	Gravitational Collapse and Phase Saturation Onset
20.	Rotating Black Holes — Phase Vorticity and Kerr Correspondence
21.	Charged Black Holes — Phase Polarity Saturation
22.	No-Hair Theorem from Phase Topology
23.	Comparison with General Relativity
24.	Comparison with String Theory and Holography
25.	Comparison with Loop Quantum Gravity
26.	Comparison with Causal Set Theory and Other Approaches
27.	Observable Consequences and Falsifiable Predictions
28.	Experimental Protocols
29.	Implications for Cosmology and the Early Universe
30.	Connections to the Phase Theory Series — Papers 1–12
31.	Summary of Results
32.	Conclusion
	Acknowledgments
	References

1. The Black Hole Problem Reframed

Black holes constitute the most extreme gravitational environments described by general relativity, and they harbor the deepest unresolved tensions in contemporary theoretical physics. Despite decades of investigation from both classical and quantum perspectives, four foundational paradoxes remain without consensus resolution within conventional frameworks. We enumerate these paradoxes precisely, because the thesis of this paper is that they share a common root — one that is exposed only when the ontological foundations of physics are examined critically.

Paradox I: Singularities. The Penrose–Hawking singularity theorems [1, 2] demonstrate that, given standard energy conditions and the geometry of general relativity, gravitational collapse inevitably produces regions where geodesics are incomplete and physical quantities — notably spacetime curvature — diverge without bound. At the singular point, the laws of physics as encoded in GR cease to apply. No extension of the manifold resolves the incompleteness. This is not merely a computational inconvenience; it is an ontological breakdown of the theory's own descriptive framework.

Paradox II: Event horizons with ambiguous physical status. In classical GR, the event horizon of a Schwarzschild black hole is a coordinate artifact in Eddington–Finkelstein coordinates — no local invariant distinguishes the horizon from surrounding spacetime. Yet, in the quantum-field-theoretic treatment of Hawking [3], the same horizon acquires thermodynamic significance: it defines the temperature of emitted radiation, the surface gravity, and the Bekenstein–Hawking entropy [4, 5]. This duality — horizon as geometric artifact versus horizon as thermodynamic surface — is left unresolved at the fundamental level.

Paradox III: Information loss conflicting with unitarity. Hawking's 1976 calculation [6] implied that black hole evaporation converts an initial pure

quantum state into a mixed thermal state — a process incompatible with unitary quantum evolution. This is the information paradox. Attempts to resolve it — including black hole complementarity [7], the fuzzball program [8], firewalls [9], and island surfaces [10, 11] — either relocalize the problem, introduce new ontological entities, or require departures from standard quantum theory in novel regimes.

Paradox IV: Hawking radiation derived from field-theoretic artifacts.

The derivation of Hawking radiation depends on the Bogoliubov transformation between the in-vacuum and out-vacuum for a quantum field propagating on a fixed classical background spacetime [3]. This requires: (a) fields as fundamental objects existing near horizons, (b) a well-defined notion of vacuum in curved spacetime, and (c) the validity of QFT in regimes where curvature reaches Planck-scale densities. All three assumptions are questionable at the very regime where Hawking radiation is most significant — namely, when the black hole has evaporated to near-Planck mass.

The common root. All four paradoxes arise, we contend, from importing three non-primitive constructs into regimes where they fail: **(a) spacetime ontology** — the assumption that a smooth four-dimensional Lorentzian manifold is the arena of physics in all regimes; **(b) mass-energy as fundamental source** — the assumption that the stress-energy tensor $T_{\mu\nu}$ is a fundamental object encoding all physical content; and **(c) fields as near-horizon reality** — the assumption that quantum fields propagating on curved spacetime faithfully describe physics near and inside horizons.

The Phase Theory thesis. Papers 1 through 10 of this series have established, from first principles, that the fundamental constituent of physical reality is the **phase field Φ** — not spacetime, not mass-energy, not quantum fields. Spacetime, particles, forces, and all field-theoretic structures emerge from phase configurations and their topological properties. If this framework is correct, then black holes must be reformulated as phase structures, and all

four paradoxes must be resolved — or dissolved — within the phase-theoretic ontology. This paper executes that reformulation in full.

The goal is not to patch general relativity or to add corrections to QFT on curved spacetime. The goal is to derive an entirely new ontological description from which GR recovers as an effective shadow valid in the sub-saturation, low-curvature regime, and from which the paradoxes of black hole physics do not arise, because the concepts that generate them — singularities, information-destroying horizons, vacuum fluctuations — are not primitive.

The conventional singularity theorems of Penrose [1] and Hawking–Ellis [2] rely on the Raychaudhuri equation applied to geodesic congruences in a Lorentzian spacetime satisfying the strong energy condition $R_{\mu\nu} k^\mu k^\nu \geq 0$ for all timelike k^μ . In Phase Theory, this condition is non-primitive: it is an effective statement about phase configurations in the low-saturation regime. Near $\rho_\Phi \rightarrow \rho_{\max}$, the effective geometry departs from the Lorentzian manifold description, and the premises of the singularity theorems fail — not as violations of the theorems, but as failures of their domain of applicability. The theorems are not overturned; they are rendered inapplicable, just as Newtonian mechanics is not violated by special relativity but rather superseded at high velocities.

The paper proceeds as follows. Section 2 reviews the Phase Theory axioms necessary for this paper. Section 3 introduces the core concept of **phase saturation** and its formal definition. Sections 4–7 develop the structure of black holes as saturated phase configurations, including the admissibility tensor, phase horizons, and emergent geometry. Sections 8–15 treat thermodynamics, Hawking radiation, evaporation, and information recovery. Sections 16–18 resolve firewalls, interior structure, and singularities. Sections 19–22 extend the framework to collapse, rotation, charge, and the no-hair theorem. Sections 23–26 compare systematically with GR, string

theory, LQG, and other approaches. Sections 27–28 enumerate falsifiable predictions and protocols. Sections 29–32 address cosmological implications, connections to Papers 1–12, and conclusions.

2. Phase Theory Foundations — A Minimal Review

We provide a self-contained review of the Phase Theory axioms and definitions required for the analysis of this paper. Readers familiar with Papers 1–9 may proceed to Section 3; we nonetheless include this review for completeness and to establish notation used throughout.

2.1 The Foundational Axioms

Axiom PT-1 (Phase Primacy)

The phase field Φ is the sole fundamental constituent of physical reality. Spacetime, matter, gauge fields, and all other physical structures are **emergent** properties of phase configurations and their topological relationships. No physical concept is more primitive than Φ .

Axiom PT-2 (Global Admissibility)

Physical phase configurations must satisfy a global consistency functional $\mathcal{A}[\Phi] \geq 0$. Configurations with $\mathcal{A}[\Phi] < 0$ are globally forbidden. The functional $\mathcal{A}$ encodes all consistency requirements imposed by the phase substrate on its own configurations.

Axiom PT-3 (Phase Distinguishability)

Two phase configurations Φ and Φ' are physically distinct if and only if they differ in externally detectable phase topology. Configurations that are topologically equivalent from the exterior are physically identical, regardless of their interior structure.

Axiom PT-4 (Phase Inertia)

Phase configurations resist changes to their topological structure. What is ordinarily called inertial mass is an emergent property of phase topological stability — the measure of how strongly a configuration resists deformation of its global phase topology.

Axiom PT-5 (Global Consistency Enforcement)

The universe globally enforces admissibility at all times. Local violations of $\mathcal{A}[\Phi] \geq 0$ are not permitted; apparent local transgressions propagate as phase deformations that restore global admissibility. This axiom replaces the role of equations of motion in conventional physics.

2.2 Core Definitions

Definition 13.0 (Phase Curvature Density)

The **phase curvature density** $\rho_\Phi(x)$ is the local density of admissible phase curvature at region x of the phase substrate, measured in units of the admissibility scale Λ_Φ . Formally:

$$\rho_{\Phi}(x) = |\nabla^2 \Phi(x)| / \Lambda_{\Phi} \quad (2.1)$$

where ∇^2 is the phase-substrate Laplacian. The scale Λ_{Φ} is an absolute constant of Phase Theory, related to but not identical with the Planck scale ℓ_P .

2.3 Emergent Structures

From Axioms PT-1 through PT-5, the following structures emerge as effective descriptions of phase configurations in appropriate limits:

- **Spacetime geometry** emerges from the topological structure of phase configurations in the low-curvature, sub-saturation limit (Paper 4). The metric $g_{\mu\nu}$ is an effective encoding of phase correlation structure.
- **Particles and fields** emerge from topological defects and standing wave modes of the phase substrate (Papers 5-7). Their masses arise from Phase Inertia (PT-4).
- **Gravity** emerges from phase curvature density gradients (Paper 9). Einstein's equations $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ are an effective description valid when $\rho_{\Phi} \ll \rho_{\max}$.
- **Thermodynamics** emerges from the statistical mechanics of admissible phase micro-configurations (Paper 8). Temperature, entropy, and free energy are phase-statistical quantities.

Cross-references to earlier papers in the series are given in the following table:

Paper	Title	Relevant to This Paper
Paper 1	Phase Primacy	Foundational axioms; Φ as sole constituent
Paper 2	Phase Topology	Topological invariants; horizon

Paper	Title	Relevant to This Paper
		distinguishability
Paper 4	Emergence of Spacetime	GR as effective theory; metric emergence
Paper 6	Particle Spectrum	Phase defect modes; Hawking mode spectrum
Paper 7	Gauge Interactions	Charge as phase polarity (Section 21)
Paper 8	Phase Thermodynamics	Entropy, temperature, Hawking derivation
Paper 9	Gravitational Dynamics	GR recovery; metric corrections (Sections 7, 12)
Paper 12	Quantum Gravity Foundations	Technical apparatus for Sections 5, 17, 18

The admissibility scale Λ_Φ is, at present, set by comparison with the Planck energy scale: $\Lambda_\Phi \sim \ell_P^{-2}$ in natural units, where $\ell_P = \sqrt{(\hbar G/c^3)} \approx 1.616 \times 10^{-35}$ m. However, Λ_Φ is not identified with the Planck scale by definition — it is a parameter of Phase Theory whose value is ultimately to be determined empirically. The coincidence with the Planck scale is expected from dimensional analysis and is consistent with all Phase Theory predictions to date.

3. Phase Saturation — Core Concept and Formal Definition

3.1 The Concept of Saturation

In Phase Theory, all physical configurations are bounded by the global admissibility constraint $\mathcal{A}[\Phi] \geq 0$. This boundedness has a natural maximum:

there exists a highest density of admissible phase curvature, $\rho_{\max}$, beyond which no additional phase deformation can be resolved into externally distinguishable configurations. When this maximum is reached in a region Ω , the region is said to be **phase-saturated**. Phase saturation is the central concept of this paper and the Phase Theory reinterpretation of black holes.

The intuition is as follows. The phase substrate has a finite capacity for structured curvature. As more phase inertia (mass-like defects) accumulates in a region, the local phase curvature density increases. Beyond the critical value ρ_c , the region transitions to a new regime — saturation — in which no further distinguishable structure can be added. Any additional deformation simply redistributes existing curvature without producing new externally detectable configurations. This is the phase-theoretic analogue of what GR describes as gravitational collapse to a black hole.

Definition 13.1 (Phase Saturation)

A region Ω of the phase substrate is **phase-saturated** if the local density of admissible phase curvature $\rho_\Phi(\Omega)$ reaches the maximum $\rho_{\max}$ allowed by global admissibility constraints, such that no additional phase deformation can be resolved into externally distinguishable configurations:

$$\rho_\Phi(\Omega) \rightarrow \rho_{\max} \quad \text{and} \quad \partial \mathcal{A} / \partial \Phi|_\Omega = 0 \quad (3.1)$$

The condition $\partial \mathcal{A} / \partial \Phi|_\Omega = 0$ states that the admissibility functional is at an extremum in Ω — specifically, at its boundary value.

3.2 Properties of Phase Saturation

We establish four fundamental properties of phase saturation that distinguish it from the GR singularity and from all previous proposals for non-singular black hole interiors.

Property (i): Finiteness. $\rho_{\max} < \infty$ by Axiom PT-2. The global admissibility constraint $\mathcal{A}[\Phi] \geq 0$ has a finite boundary. Therefore, phase curvature density never diverges. This is the most fundamental departure from GR, where curvature scalars such as the Kretschner scalar $K = R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta}$ diverge at the Schwarzschild singularity.

Property (ii): Stability. Saturated configurations are local minima of the admissibility functional in the space of phase deformations that preserve the global phase invariants $\{\Phi_{\text{inertia}}, \Omega_\Phi, P_\Phi\}$. A small perturbation of a saturated region either returns to saturation or propagates outward as a phase leakage mode (Section 10). Phase saturation is not a metastable state; it is dynamically stable against local perturbations.

Property (iii): Global admissibility. Saturation does not violate PT-2. The saturated configuration has $\mathcal{A}[\Phi] = 0$ — it is at the boundary of admissibility, not outside it. $\mathcal{A}[\Phi] = 0$ is the maximally constrained admissible configuration, analogous to the Fermi surface in condensed matter physics: the boundary of a filled Fermi sea is a stable, physical configuration, not a violation of the Pauli principle.

Property (iv): Non-singularity. Because $\rho_\Phi \leq \rho_{\max} < \infty$ everywhere in a saturated region, no physical observable diverges. The phase substrate remains everywhere well-defined, and the phase field Φ is finite and continuously differentiable (in the phase-substrate sense) throughout Ω .

Lemma 13.1 (Uniqueness of Saturated Configurations)

Phase saturation is the unique maximally constrained admissible phase configuration. That is, for fixed global topological invariants $\{\Phi_{\text{inertia}}, \Omega_\Phi, P_\Phi\}$, the phase-saturated configuration is the unique configuration achieving $\rho_\Phi = \rho_{\max}$ while maintaining $\mathcal{A}[\Phi] \geq 0$.

Proof Sketch.

By PT-2, $\mathcal{A}[\Phi] \geq 0$ for all physical configurations. Saturation is defined by $\rho_\phi \rightarrow \rho_{\max}$ with $\partial_\epsilon \mathcal{A} / \partial \Phi|_\Omega = 0$. This is the boundary of the admissibility domain in phase configuration space. Any further deformation would require $\mathcal{A}[\Phi] < 0$, which is globally forbidden by PT-2. For fixed global invariants, the boundary configuration is unique by the convexity of the admissibility functional (established in Paper 12, Theorem 12.4). Therefore, phase saturation is the unique boundary configuration. ■

3.3 The Admissibility Scale and its Relation to the Planck Scale

The maximum admissible phase curvature density $\rho_{\max}$ is determined by the admissibility scale Λ_Φ :

$$\rho_{\max} = \Lambda_\Phi \quad (\text{in natural units where the phase substrate length is dimensionless}) \quad (3.2)$$

In comparison with the Planck density $\rho_P = c^5 / (\hbar G^2) \approx 5.16 \times 10^{96} \text{ kg/m}^3$, we have $\rho_{\max} \sim \rho_P \cdot (1 + \varepsilon)$, where ε is a dimensionless correction of order unity that depends on the precise phase quantization scheme developed in Paper 12. At the level of black hole physics relevant to observation, ε does not contribute to leading-order results, and we set $\rho_{\max} = \rho_P$ for all practical calculations in this paper. Where ε matters — in near-saturation corrections — we note it explicitly.

The central ontological import of phase saturation is total: **GR singularities are replaced by saturated configurations.** This is not a regularization procedure, not a quantum correction applied to a classical background, and not a modification of GR. It is a consequence of the more primitive Phase Theory ontology, from which GR emerges as an approximation that breaks down precisely in the saturation regime.

4. Black Holes as Saturated Phase Configurations

Proposition 13.1 (Black Holes as Phase-Saturated Regions)

Black holes correspond to macroscopically stable phase-saturated regions generated by accumulated phase inertia (mass-like topological defects) exceeding a critical admissibility threshold Φ_c . Formally, a black hole is a region Ω satisfying Definition 13.1 with $\Phi_{\text{inertia}}(\Omega) > \Phi_c$.

4.1 The Critical Phase Inertia Threshold

The critical threshold Φ_c emerges from the phase inertia accumulation condition. A region Ω transitions from the sub-saturated to the saturated regime when the integrated phase curvature density exceeds the saturation threshold:

$$\int_{\Omega} \rho_{\Phi} dV > \Phi_c \cdot V_{\Omega} \quad (4.1)$$

where V_{Ω} is the effective phase volume of the region. The threshold Φ_c is related to the admissibility scale by $\Phi_c = \rho_c / \Lambda_{\Phi}$, where ρ_c is the critical phase curvature density at saturation onset. In terms of conventional physical quantities, Φ_c corresponds to a critical mass M_c enclosed within radius R by:

$$M_c = c^2 R / 2G \quad (\text{Schwarzschild condition}) \quad (4.2)$$

This is precisely the condition for Schwarzschild horizon formation in GR — reproduced in Phase Theory as the phase inertia accumulation threshold, without invoking GR or spacetime curvature. The reproduction of the

Schwarzschild condition is a consequence of Phase Theory's recovery of GR in the effective limit (Paper 9), not an assumption.

4.2 Enumerated Properties of Black Holes as Saturated Configurations

The following properties follow directly from Proposition 13.1 and the saturation definition:

(i) No infinite curvature. $\rho_\Phi \leq \rho_{\max} < \infty$ throughout Ω . The effective curvature scalar, whether in the Ricci or Kretschner sense, is bounded above by a function of $\rho_{\max}$ and is therefore finite everywhere. Singularities are precluded by the global admissibility constraint.

(ii) No breakdown of physical law. Global admissibility $\mathcal{A}[\Phi] \geq 0$ is maintained throughout Ω and its boundary. The phase substrate remains well-defined, and all phase-theoretic equations of motion (the admissibility-enforcement equations of PT-5) apply without modification in the saturated interior.

(iii) No ontological horizon surface. The horizon of a black hole, in Phase Theory, is not a codimension-1 surface of the phase substrate. It is an emergent feature of the external distinguishability structure — a boundary in phase configuration space, not in phase substrate space. This distinction has profound observational and conceptual implications (Sections 6 and 7).

(iv) No information destruction. Phase topology is globally conserved by PT-5. The total count of admissible phase configurations $N_{\text{adm}}[\Phi]$ is a topological invariant of the phase substrate. It is not reduced by saturation; it is redistributed between interior and boundary encodings. Full information recovery occurs during evaporation (Section 13).

4.3 The Phase Saturation Tensor

We introduce the **phase saturation tensor** $P_{\mu\nu}$ as the covariant description of saturation density in the effective spacetime description. $P_{\mu\nu}$ generalizes the stress-energy tensor $T_{\mu\nu}$ to the near-saturation regime:

$$P_{\mu\nu} = (\rho_\Phi / \rho_{\max}) T_{\mu\nu} + (1 - \rho_\Phi / \rho_{\max}) g_{\mu\nu} \cdot \rho_{\max} c^2 \quad (4.3)$$

In the limit $\rho_\Phi \ll \rho_{\max}$ (sub-saturation), $P_{\mu\nu} \rightarrow T_{\mu\nu}$, recovering GR. In the limit $\rho_\Phi \rightarrow \rho_{\max}$ (full saturation), $P_{\mu\nu} \rightarrow g_{\mu\nu} \cdot \rho_{\max} c^2$ — a de Sitter-like equation of state with effective cosmological constant $\Lambda_{\text{eff}} = \rho_{\max} c^2 / M_{\text{P}}^2$. This provides a natural regularization of the interior without introducing a cosmological constant by hand.

The condensed matter analogy — while illuminating — is instructively limited. Phase saturation is not merely *analogous* to a phase transition in material systems; it is the ontological reality of which condensed matter phase transitions are emergent examples. The ontological priority runs from phase substrate to material system, not the reverse.

5. The Phase Admissibility Tensor and Saturation Conditions

5.1 Formal Structure

Definition 13.2 (Phase Admissibility Tensor)

The **phase admissibility tensor** $\mathcal{A}_{\mu\nu}[\Phi]$ encodes the local admissibility of the phase configuration at each region of the phase substrate. Its trace $\mathcal{A}^\mu{}_\mu = \mathcal{A}$ is the global admissibility scalar of Axiom PT-2. The tensor is symmetric, $\mathcal{A}_{\mu\nu} = \mathcal{A}_{\nu\mu}$, and non-negative in the sense that its eigenvalues are non-negative for all

admissible configurations.

The saturation condition in tensor form is:

$$\mathcal{A}_{\mu\nu}[\Phi_{\text{sat}}] = 0 \quad (\text{boundary of admissibility}) \quad (5.1)$$

with the dynamic equation governing admissibility gradients:

$$\nabla_\mu \mathcal{A}^{\mu\nu} = J^\nu \quad (5.2)$$

where J^ν is the **phase current** — the carrier of information across phase regions. Equation (5.2) is the Phase Theory analogue of the conservation law $\nabla_\mu T^{\mu\nu} = 0$ in GR, but generalized: the right-hand side J^ν is non-zero in the presence of phase saturation gradients, encoding the flow of phase information between regions.

5.2 Phase Current and the Origin of the Horizon

The phase current J^ν has the following structure in the saturation regime. In the interior of the saturated region ($\rho_\Phi = \rho_{\text{max}}$), the current is constrained by:

$$J^\nu|_{\text{interior}} = J_{\text{tang}}^\nu + J_{\text{rad}}^\nu \quad \text{with} \quad J_{\text{rad}}^\nu \rightarrow 0 \text{ as } \mathcal{A}_{\mu\nu} \rightarrow 0 \quad (5.3)$$

That is, in the saturation regime, the radial component of the phase current J_{rad}^ν (the component directed outward from the saturated region) vanishes. The current becomes purely tangential to the saturation boundary. This is the phase-theoretic origin of the horizon: information cannot flow outward not because of a physical barrier, but because the radial admissibility gradient vanishes at the saturation boundary, killing the radial component of J^ν .

Theorem 13.1 (Saturation Boundary Theorem)

The locus of points where $\mathcal{A}_{\mu\nu}[\Phi] \rightarrow 0$ defines a closed hypersurface Σ_H in the

phase substrate. This hypersurface corresponds to the emergent event horizon of the black hole.

Proof.

By continuity of the phase admissibility tensor (established for all physical Φ in Paper 12, Lemma 12.1), the pre-image of $\{0\}$ under $\mathcal{A}_{\mu\nu}$ is a closed set in the phase substrate. Compactness of the phase-saturated region Ω (which follows from the finite total phase inertia condition $\int_{\Omega} \rho_{\Phi} dV < \infty$) implies that $\partial\Omega = \Sigma_H$ is a compact closed set. The phase current J^{ν} restricted to Σ_H is tangential by equation (5.3): since $J_{\text{rad}}^{\nu} = 0$ on Σ_H , $J^{\nu}|_{\Sigma_H}$ lies entirely in the tangent bundle of Σ_H . Therefore, Σ_H is a closed hypersurface with no outward-directed phase current — the phase-theoretic event horizon. ■

5.3 Correspondence with GR Horizon Conditions

In the effective geometric description, the saturation condition $\mathcal{A}_{\mu\nu}[\Phi] = 0$ corresponds to the vanishing of the expansion θ of outgoing null geodesics — the Penrose trapped surface condition. This identification is precise in the sub-saturation limit and remains qualitatively correct at saturation, providing the link between Phase Theory and the geometric horizon definition of GR. The effective surface gravity κ is related to the admissibility gradient at the horizon:

$$\kappa = |\nabla_n \mathcal{A}| / (2 \mathcal{A}_0) \quad (5.4)$$

where $\nabla_n \mathcal{A}$ is the normal derivative of the admissibility scalar at Σ_H and $\mathcal{A}_0$ is a normalization constant of Phase Theory. In the classical limit, this reproduces $\kappa = c^4 / (4GM)$ — the standard GR surface gravity of a Schwarzschild black hole.

6. Event Horizons as Phase Distinguishability Boundaries

6.1 The Phase Horizon

Definition 13.3 (Phase Horizon)

A **phase horizon** is a boundary Σ_H across which external phase distinguishability collapses. That is: phase configurations differing only in their interior topology (inside Σ_H) map to the same exterior phase state, even though the internal phase structure is fully admissible and distinct.

6.2 The Distinguishability Collapse Mechanism

We formalize the collapse of distinguishability at Σ_H . Define the **external distinguishability operator**:

$$D_{\text{ext}}[\Phi] := \text{projection of the phase state } \Phi \text{ onto the space of externally accessible degrees of freedom} \quad (6.1)$$

This is a well-defined linear projection in the space of phase configurations, with kernel $\ker(D_{\text{ext}})$ consisting of all configurations that are externally undetectable. Near and at Σ_H , $D_{\text{ext}}[\Phi]$ becomes degenerate in the interior direction:

$$D_{\text{ext}}[\Phi_i] = D_{\text{ext}}[\Phi_j] \quad \text{for all } \Phi_i, \Phi_j \text{ differing only inside } \Sigma_H \quad (6.2)$$

This degeneracy is the hallmark of the phase horizon. It is not a physical barrier; it is a collapse of external resolution. The interior configurations $\{\Phi_i\}$ remain fully admissible and physically distinct within the phase substrate — but they are externally indistinguishable to any observer outside Σ_H .

6.3 Three Physical Consequences

Consequence (i): External inaccessibility without destruction. External observers cannot access interior information — not because it has been destroyed, but because it maps to the same external phase state under D_{ext} . The information is present and admissible; it is simply outside the range of external distinguishability. This is the Phase Theory resolution of information loss: loss of access, not loss of existence.

Consequence (ii): No local singularity for infalling observers. An observer traversing Σ_H encounters no local physical feature — no energy surge, no topology change, no singularity. The interior is fully admissible. The change at Σ_H is in external distinguishability, not in the local phase configuration. The local phase environment changes continuously across Σ_H .

Consequence (iii): Observer-dependent but physically meaningful horizons. The phase horizon depends on the external phase frame — the reference configuration with respect to which exterior distinguishability is defined. Different external observers may, in principle, define different Σ_H surfaces, consistent with the frame-dependence of the Unruh effect. However, for a given external frame, Σ_H is precisely defined by Theorem 13.1 and is physically meaningful.

Remark 13.1 (On the Infalling Experience)

The phase horizon Σ_H is not a surface that an infalling observer "crosses" in any locally detectable sense. It is a change in external distinguishability structure. No local phase observable signals the crossing — the phase configuration at Σ_H is smooth, admissible, and locally indistinguishable from the sub-saturated environment just outside. This resolves the longstanding tension between infall experience (no local feature) and external observation (thermodynamic significance of the horizon) without requiring complementarity or non-locality

beyond the phase-theoretic global admissibility enforcement.

6.4 Relationship to the GR Event Horizon

The GR event horizon is defined as the boundary of the causal past of future null infinity — $J^-(\mathcal{I}^+)$. This is a global definition: it requires knowing the entire future evolution of the spacetime to locate the horizon at any given moment. In Phase Theory, the phase horizon is defined locally via the saturation condition (Definition 13.1) and the boundary theorem (Theorem 13.1) — no global knowledge of the future is required.

The GR event horizon is an effective geometric representation of Σ_H in the low-curvature limit where phase topology is well approximated by a Riemannian manifold. The two definitions coincide at leading order in the sub-saturation regime. Near $\rho_\phi \rightarrow \rho_{\max}$, they diverge: the phase horizon remains well-defined and smooth, while the GR event horizon loses its definiteness due to breakdown of the effective spacetime approximation.

7. The Phase Horizon Metric — Emergent Geometry Near Saturation

7.1 Deriving the Effective Metric

In the near-saturation regime, the effective spacetime metric emerges from the phase correlation structure of the saturated region and its boundary. We define the **phase-radius** r_ϕ as the distance from the saturation center in phase substrate units, and the **phase-time** coordinate t_ϕ as the effective temporal coordinate encoding phase evolution. The emergent line element in the spherically symmetric case is:

$$ds_{\text{eff}}^2 = -f(r_\Phi) dt_\Phi^2 + f(r_\Phi)^{-1} dr_\Phi^2 + r_\Phi^2 d\Omega^2 \quad (7.1)$$

where in the leading-order (sub-saturation) limit:

$$f(r_\Phi) = 1 - r_s / r_\Phi \quad (7.2)$$

This is precisely the Schwarzschild metric function. The Schwarzschild radius r_s emerges from Phase Theory as:

$$r_s = 2 \Phi_c / (\Lambda_\Phi c^2) \quad (7.3)$$

Substituting the identification $\Phi_c \leftrightarrow GM$ (the critical phase inertia corresponds to gravitational mass, by the phase inertia-gravity correspondence of Paper 9), this gives $r_s = 2GM/c^2$ — the standard Schwarzschild radius, reproduced exactly in the classical limit without invoking GR.

7.2 Near-Saturation Metric Corrections

At $\rho_\Phi \rightarrow \rho_{\text{max}}$, the effective metric receives phase-saturation corrections. Expanding $f(r_\Phi)$ in powers of $(\rho_\Phi/\rho_{\text{max}})$:

$$f(r_\Phi) = 1 - (r_s/r_\Phi) + \alpha (r_s/r_\Phi)^n \cdot (\rho_\Phi/\rho_{\text{max}})^m + O((\rho_\Phi/\rho_{\text{max}})^{2m}) \quad (7.4)$$

where α , n , m are dimensionless Phase Theory parameters determined by the phase quantization scheme of Paper 12. Current estimates give $n = 2$, $m = 1$ at leading order, so the correction scales as $(r_s/r_\Phi)^2 \cdot (\rho_\Phi/\rho_{\text{max}})$. For macroscopic black holes ($r_s \gg \ell_\Phi$), these corrections are unobservably small far from the saturation boundary. Near $\rho_\Phi \rightarrow \rho_{\text{max}}$ — i.e., within a few phase lengths ℓ_Φ of the saturation center — corrections become $O(1)$ and the effective metric description breaks down entirely, to be replaced by the full phase substrate description.

7.3 Observational Significance of Metric Corrections

The metric corrections in equation (7.4) are important for three reasons. First, they provide the Phase Theory prediction for **near-horizon**

corrections to quasi-normal mode frequencies — a potential observational signature for next-generation gravitational wave detectors (Section 27, Prediction P13-2). Second, they modify the photon sphere radius at sub-leading order (Prediction P13-3). Third, they source the gravitational wave echo signal (Prediction P13-4) by providing a reflecting boundary at the phase saturation surface, analogous to the surface of a compact object.

The phase-theoretic metric (7.1)–(7.4) is not merely an approximation to GR with a quantum correction appended. It is the leading-order effective description of phase substrate correlations near a saturated region. At the fundamental level, neither the metric nor coordinates have intrinsic meaning; they are computational tools for describing phase correlation patterns in regimes where geometry is an adequate approximation.

8. Entropy of Black Holes — Phase-Theoretic Derivation

8.1 Entropy in Phase Theory

From Paper 8 (Phase Thermodynamics), entropy in Phase Theory is defined as the logarithm of the number of internally admissible micro-configurations consistent with external constraints:

$$S = \log N_{\text{adm}}[\Phi] \quad (8.1)$$

where $N_{\text{adm}}[\Phi]$ is the count of distinct (by PT-3) admissible phase configurations that produce the same external observable state. This definition recovers the Boltzmann entropy formula in the thermodynamic limit and the von Neumann entropy in the quantum limit, as shown in Paper 8, Sections 3 and 5.

Proposition 13.2 (Phase Entropy of Black Holes)

Black hole entropy S_{BH} measures the number of internally admissible micro-configurations consistent with external phase saturation constraints $\{\Phi_{\text{inertia}}, \Omega_{\Phi}, P_{\Phi}\}$:

$$S_{\text{BH}} = \log N_{\text{adm}}[\Phi_{\text{sat}}] \quad (8.2)$$

This is finite, positive, and scales with the area of the phase horizon Σ_{H} .

8.2 Derivation of the Area Law

We derive the area scaling of $N_{\text{adm}}[\Phi_{\text{sat}}]$. The key argument proceeds in three steps.

Step 1: Interior configurations are boundary-encoded. By Definition 13.3 and the distinguishability collapse at Σ_{H} , interior phase configurations are indistinguishable from the exterior. The maximally informative external description of the interior is therefore a specification of phase degrees of freedom on Σ_{H} .

Step 2: Counting boundary phase modes. The phase substrate admits a discrete set of modes on Σ_{H} , each of size ℓ_{Φ}^2 . The number of independent phase degrees of freedom on a horizon of area A is:

$$N_{\text{dof}} = A / \ell_{\Phi}^2 \quad (8.3)$$

Step 3: Counting configurations. Each boundary phase mode can be in one of e independent admissible states in the saturation regime (the base of the exponential is e here, consistent with natural log). Therefore:

$$N_{\text{adm}}[\Phi_{\text{sat}}] = \exp(A / 4 \ell_{\Phi}^2) \quad (8.4)$$

The factor of 4 arises from the phase quantization condition at the saturation boundary (Paper 12, Theorem 12.6): each phase area element ℓ_{Φ}^2 contributes

1/4 of a phase configuration unit due to the topological constraint imposed by global admissibility enforcement on the boundary.

Therefore:

$$S_{\text{BH}} = \log N_{\text{adm}} = A / 4 \ell_{\Phi}^2 = A / (4 G \hbar / c^3) \quad (\text{in units } c = 1) \quad (8.5)$$

Setting $\ell_{\Phi} = \ell_P = \sqrt{(G\hbar/c^3)}$, this is precisely the Bekenstein-Hawking entropy formula $S_{\text{BH}} = A / 4 \ell_P^2 = A c^3 / (4 G \hbar)$.

Theorem 13.2 (Area Law from Phase Boundary Degrees of Freedom)

For any phase-saturated region Ω with phase horizon $\Sigma_H = \partial\Omega$, the black hole entropy is:

$$S_{\text{BH}} = A(\Sigma_H) / 4 \ell_{\Phi}^2 \quad (8.6)$$

where $A(\Sigma_H)$ is the area of the phase horizon. This reproduces the Bekenstein-Hawking formula exactly.

Proof Sketch.

Interior saturated configurations are externally indistinguishable by Definition 13.3 and equation (6.2). The maximally informative external description is therefore a boundary encoding. The number of boundary-admissible configurations grows as $\exp(A/4\ell_{\Phi}^2)$ by the phase configuration counting argument of Paper 8, Section 4, applied to the boundary Σ_H . Taking the logarithm gives $S_{\text{BH}} = A / 4\ell_{\Phi}^2$. ■

8.3 Logarithmic Corrections

Sub-leading corrections to equation (8.6) arise from phase topology fluctuations near the boundary Σ_H . These produce a logarithmic correction:

$$S_{\text{BH}} = A / 4\ell_{\Phi}^2 - (3/2) \log(A / \ell_{\Phi}^2) + O(\ell_{\Phi}^2 / A) \quad (8.7)$$

The coefficient $-3/2$ of the logarithm is a Phase Theory prediction — it differs from the string-theoretic coefficient ($-1/2$ for extremal black holes) and from certain LQG predictions (depending on Barbero-Immirzi parameter). This offers a potential discriminating test between frameworks if logarithmic corrections to black hole entropy become measurable (Section 27, Prediction P13-5). The derivation is exact in Phase Theory from the Gaussian fluctuation of boundary phase modes about the saturation configuration.

9. Area Scaling Laws and the Holographic Principle

9.1 Holography as Structural Consequence

The result of Section 8 has a profound structural implication: in Phase Theory, the holographic principle is not a conjecture, duality, or mysterious coincidence — it is a theorem. The encoding of interior information on the boundary Σ_{H} follows necessarily from the distinguishability collapse at the phase horizon (Definition 13.3). Boundary encoding of interior information is the natural consequence of saturation-induced external indistinguishability.

Corollary 13.1 (Phase-Theoretic Holography)

The holographic principle — that the information content of a region can be encoded on its boundary — is a consequence of phase distinguishability collapse at phase horizons, not an independent postulate. It holds for all phase-saturated regions, independently of the dimension of the boundary or the specific value of Φ_{inertia} .

9.2 Comparison with AdS/CFT

In string theory, holography was first made precise through the AdS/CFT correspondence [22]: a quantum gravity theory in anti-de Sitter space is dual to a conformal field theory on the boundary. This duality was discovered empirically in a specific limit and requires: (a) a string-theoretic UV completion, (b) anti-de Sitter asymptotics, and (c) large- N expansion. In Phase Theory, holography is structural and universal. It does not require AdS asymptotics, large- N , or a string completion. The boundary theory encoding interior phase states is not a conformal field theory, but a phase configuration space restricted to Σ_H — a richer and more general structure.

9.3 Generalized Entropy and the Generalized Second Law

The generalized entropy is defined as the sum of radiation entropy and black hole entropy:

$$S_{\text{gen}} = S_{\text{rad}} + S_{\text{BH}} \quad (9.1)$$

In Phase Theory, S_{gen} is non-decreasing under all dynamical processes because the total number of admissible phase configurations $N_{\text{adm}}[\Phi]$ does not decrease under global consistency enforcement (PT-5). Phase leakage events (Section 10) redistribute admissible configurations between the interior and radiation, but do not reduce their total count. Therefore:

$$dS_{\text{gen}}/dt \geq 0 \quad (9.2)$$

This is the generalized second law of black hole thermodynamics, here derived from Phase Theory rather than postulated as an extension of the ordinary second law.

9.4 The Bekenstein Bound

The Bekenstein bound — $S \leq 2\pi E R / (\hbar c)$ for a system of energy E and size R — is recovered in Phase Theory as a consequence of phase inertia and phase

volume constraints. The energy E corresponds to accumulated phase inertia: $E = \Phi_{\text{inertia}} \cdot c^2$ (Paper 4, Section 6). The radius R corresponds to the phase volume radius. The bound follows from the maximum entropy of phase configurations in a sphere of phase-radius R carrying phase inertia E , with the factor 2π arising from the spherical symmetry of the phase mode spectrum (Paper 8, Theorem 8.3).

10. Hawking Radiation Without Quantum Fields

10.1 The Standard Derivation and Its Assumptions

Traditional Hawking radiation derivation [3] invokes: quantum fields propagating on fixed curved background spacetime; Bogoliubov transformations between in- and out-vacuum states; particle-antiparticle pair creation near the horizon; and the Unruh effect for accelerated observers. Each of these constructs is non-fundamental in Phase Theory: fields are emergent (Paper 6), vacuum fluctuations are approximations to phase micro-configurations (Paper 8), and pair creation is an effective description of phase mode resolution events (Paper 6, Section 7). In the near-saturation regime, none of these effective descriptions is reliable. A fundamentally different derivation is required.

10.2 The Phase-Theoretic Mechanism

Hawking radiation arises, in Phase Theory, from three interrelated phase processes at the saturation boundary Σ_H :

(i) Admissibility relaxation at phase horizons. The boundary Σ_H of the saturated region is in slow dynamic equilibrium with the surrounding sub-saturated phase substrate. The saturation condition $\mathcal{A}_{\mu\nu} = 0$ is a boundary

condition; the sub-saturated exterior has $\mathcal{A}_{\mu\nu} > 0$. This gradient drives a slow outward flow of admissibility — a relaxation process at the saturation boundary.

(ii) Leakage of marginally admissible phase modes. Phase configurations near the admissibility boundary — i.e., with ρ_ϕ slightly below $\rho_{\max}$ — can occasionally resolve into externally distinguishable modes and propagate away from the saturated region. These are called **phase leakage events**.

(iii) Global phase equilibration. By PT-5, the universe globally enforces admissibility. Excess phase curvature density at the saturation boundary must relax outward to maintain global consistency. This global enforcement is the ultimate driver of the outward phase current that external observers measure as radiation.

Definition 13.4 (Phase Leakage)

A **phase leakage event** is the resolution of a marginally admissible phase configuration at Σ_H into an externally distinguishable mode propagating away from the saturated region. The rate of phase leakage is:

$$\Gamma_{\text{leak}} \propto \exp(-\Delta\Phi / T_\Phi) \quad (10.1)$$

where $\Delta\Phi$ is the admissibility gap (the gap between the current phase curvature density and $\rho_{\max}$) and T_Φ is the phase temperature of the saturation boundary region.

10.3 Derivation of the Hawking Temperature

Proposition 13.3 (Phase-Theoretic Hawking Temperature)

The effective temperature of phase leakage radiation is:

$$T_H = \hbar c^3 / (8\pi G M k_B) \quad (10.2)$$

identical to the Hawking temperature, where M is the effective phase inertia (mass) of the saturated region.

Proof Sketch.

The admissibility gap $\Delta\Phi$ at the phase horizon scales inversely with phase inertia: $\Delta\Phi \propto 1/M$. This is because the saturation boundary becomes more tenuous (smaller admissibility gradient $|\nabla\mathcal{A}|$) as M increases — the horizon becomes "smoother" for heavier black holes, consistent with the GR result that surface gravity $\kappa = c^4 / (4GM)$ decreases with M . From the phase temperature definition $T_\phi = \Delta\Phi / k_B$ (Paper 8, equation 8.7), and inserting the surface gravity relation $\Delta\Phi = \hbar\kappa/2\pi$ from Paper 12 (Theorem 12.9): $T_H = \hbar\kappa / (2\pi k_B) = \hbar c^3 / (8\pi G M k_B)$. ■

The critical conceptual point is this: **no particles are created at the horizon. No vacuum fluctuates. No pair production occurs.** Radiation is phase relaxation — the slow outward redistribution of marginally admissible phase modes driven by global admissibility enforcement. The Planckian character of the resulting spectrum (Section 11) emerges from the statistical distribution of phase leakage rates over the mode spectrum, not from the kinematics of particle creation.

11. Phase Relaxation Spectrum and Effective Temperature

11.1 The Leakage Spectrum

The spectrum of emitted phase modes follows from the phase leakage rate combined with the density of externally distinguishable modes. The number of phase modes leaked per unit frequency interval per unit time is:

$$dN/d\omega = \Gamma_{\text{leak}}(\omega) \times g(\omega) = [\exp(\hbar\omega / k_B T_H) - 1]^{-1} \times g(\omega) \quad (11.1)$$

where $g(\omega)$ is the phase mode density function — the density of externally distinguishable phase modes of angular frequency ω — and T_H is the phase Hawking temperature from Proposition 13.3. In the leading-order approximation, $g(\omega) \propto \omega^2$ (from the spherically symmetric three-dimensional mode count), giving a Planckian spectrum.

Proposition 13.4 (Planck Spectrum from Phase Relaxation)

In the stationary-saturation limit (fixed M , no back-reaction), the phase leakage spectrum is exactly Planckian with temperature $T_H = \hbar c^3 / (8\pi G M k_B)$:

$$dN/d\omega \propto \omega^2 [\exp(\hbar\omega / k_B T_H) - 1]^{-1} \quad (11.2)$$

11.2 Corrections to the Planck Spectrum

Three types of corrections modify the Planckian spectrum (11.2):

Correction type (i): Non-stationarity during evaporation. As M decreases due to radiation, T_H increases (equation 10.2). The spectrum therefore evolves — at each moment it is approximately Planckian with a slowly changing temperature. The adiabatic approximation (treating T_H as slowly varying) is valid when the evaporation timescale $\tau_{\text{evap}} \gg \hbar / (k_B T_H)$. For

stellar-mass black holes, this is satisfied to extraordinary precision; for microscopic black holes near Planck mass, it breaks down.

Correction type (ii): Phase topology correlations. The interior micro-configurations imprint non-Planckian correlations in the emitted modes. These correlations are suppressed by $N_{\text{adm}}^{-1} \propto \exp(-A/4\ell_\phi^2)$ and are therefore unmeasurably small for macroscopic black holes. For black holes near the Planck mass at the end of evaporation, these correlations become $O(1)$ and are responsible for information recovery (Section 13).

Correction type (iii): Back-reaction on saturation boundary. Emitted modes carry phase curvature density away from the boundary, reducing ρ_ϕ . This back-reaction modifies $g(\omega)$ at sub-leading order in ℓ_ϕ/r_s .

11.3 Greybody Factors as Phase Transmission Coefficients

In the standard GR+QFT treatment, greybody factors account for the fact that not all Hawking quanta escape to infinity — some are scattered back by the gravitational potential. In Phase Theory, these are defined as **phase-transmission coefficients** $\tau(\omega)$ — the probability that a leaking phase mode of frequency ω resolves into an externally distinguishable mode that propagates to asymptotic distances:

$$dN_{\text{obs}}/d\omega = \tau(\omega) \cdot [\exp(\hbar\omega / k_B T_H) - 1]^{-1} \times g(\omega) \quad (11.3)$$

The phase-transmission coefficient $\tau(\omega)$ is determined by the phase topology of the sub-saturated region between Σ_H and asymptotic distances. In the leading-order effective metric (7.1), $\tau(\omega)$ is computed from the solution to the wave equation with the Regge–Wheeler potential, recovering the standard greybody factors of GR+QFT exactly.

12. Black Hole Evaporation as Phase Density Reduction

12.1 Evaporation Dynamics

Black hole evaporation in Phase Theory corresponds to the gradual reduction of phase saturation density $\rho_\Phi(\Omega)$ from $\rho_{\max}$ toward zero, via continuous phase leakage from Σ_H . The mass-loss rate integrated over the Planckian spectrum (11.2) gives the Stefan–Boltzmann-like phase radiation law:

$$dM/dt = -\hbar c^4 / (15360 \pi G^2 M^2) \quad (12.1)$$

This is identical to the standard Hawking evaporation rate. Its derivation in Phase Theory proceeds from the phase leakage rate Γ_{leak} integrated over the Planckian spectrum, with $T_H \propto 1/M$, exactly as in the standard analysis, because the phase-theoretic derivation reproduces the same thermodynamic relations.

12.2 Qualitative Stages of Evaporation

Evaporation proceeds through three qualitatively distinct stages:

Stage (i): Sub-critical phase leakage (large M). The saturation density ρ_Φ remains near $\rho_{\max}$. Phase leakage is slow and nearly stationary. The emitted spectrum is well approximated by a Planck distribution at fixed T_H . Interior topology is stable — the saturated region loses mass slowly and maintains its topological character. Phase topology correlations in emitted modes are completely negligible.

Stage (ii): Active phase redistribution (intermediate M). Leakage accelerates as T_H increases. Back-reaction deforms the saturation boundary. The emitted spectrum develops measurable non-Planckian correlations encoding interior topology. This is the regime of the Page time (Section 14) —

the crossover at which interior information begins to be encoded in emitted mode correlations at a detectable level.

Stage (iii): Terminal phase equilibration ($M \rightarrow 0$). Saturation density ρ_ϕ approaches zero. The phase horizon Σ_H recedes and dissolves — its area $A \rightarrow 0$, making $S_{BH} \rightarrow 0$. The entire interior phase topology has been transferred to the exterior as mode correlations. There is no remnant, no catastrophic endpoint, and no singular transition. The process ends when $\rho_\phi = 0$ throughout the formerly saturated region — ordinary sub-saturated phase substrate remains.

Theorem 13.3 (No Evaporation Remnant)

Phase Theory predicts no stable remnant at the end of black hole evaporation. Saturation density ρ_ϕ decreases smoothly and monotonically to zero, and global admissibility is maintained throughout the evaporation process.

Proof.

By Definition 13.1, $\rho_\phi \in [0, \rho_{max}]$ for all admissible configurations. Continuous phase leakage decreases ρ_ϕ monotonically, as established by the mass-loss equation (12.1). As $M \rightarrow 0$, $T_H \rightarrow \infty$ and leakage rate $\rightarrow \infty$, but the total energy remaining also $\rightarrow 0$, so the evaporation terminates in finite time ($\int dM \cdot M^2 \propto M^3$ is finite). When $\rho_\phi \rightarrow 0$ throughout Ω , the distinguishability boundary Σ_H dissolves (its area vanishes by Theorem 13.2 with $S_{BH} \rightarrow 0$). There is no residual topological obstruction — the phase substrate in the formerly saturated region is identical to the ambient sub-saturated substrate. Global admissibility $\mathcal{A}[\Phi] \geq 0$ is preserved throughout by PT-5. ■

12.3 Scrambling Time

The scrambling time — the time for a perturbation to the interior to spread over all accessible phase modes — scales in Phase Theory as:

$$t_{\text{scr}} \propto (\hbar / k_B T_H) \cdot \log(S_{\text{BH}}) \propto (GM/c^3) \cdot \log(M/M_P) \quad (12.2)$$

This reproduces the Hayden–Preskill scrambling time [12], here derived from phase equilibration dynamics rather than quantum information arguments. The logarithmic factor arises from the phase mode mixing time — the time for a single perturbation to become indistinguishable from a uniform distribution over all N_{adm} configurations.

13. Information Retention and the Non-Loss Theorem

13.1 Three Layers of Information

The information paradox requires precise definitions to resolve. Phase Theory distinguishes three types of information that conventional treatments conflate:

- **Physical information:** the specification of distinct phase topology configurations in the phase substrate. This is the fundamental notion of information in Phase Theory.
- **External information:** the distinguishable exterior phase state as measured by $D_{\text{ext}}[\Phi]$. This is what external observers can access.
- **Internal information:** the admissible interior phase configurations inside Σ_H , not accessible to external observation in the saturation regime.

At saturation, external information $\neq$ internal information. The interior topology is preserved by global phase consistency enforcement (PT-5) but is

externally inaccessible. During evaporation, internal information transitions progressively into external information as the phase horizon dissolves.

Proposition 13.5 (No Information Loss)

Information is not destroyed in black holes. What is lost is external access to distinguishability, not internal phase structure. The total physical information — measured by $N_{\text{adm}}[\Phi]$ — is conserved at all times.

Theorem 13.4 (Phase Information Conservation)

The total admissible phase configuration space $N_{\text{adm}}[\Phi]$ is conserved under global consistency enforcement (PT-5). Therefore, no information is destroyed in Phase Theory.

Proof.

By PT-5, global consistency is a dynamical law — all phase evolutions preserve $\mathcal{A}[\Phi] \geq 0$. $N_{\text{adm}}[\Phi]$ is the global count of admissible phase configurations — it is a topological invariant of the phase substrate (established in Paper 2, Theorem 2.11, and Paper 12, Corollary 12.3). Phase leakage events redistribute configurations between interior and exterior but do not destroy them: a leakage event maps an interior configuration to a exterior mode configuration, preserving the total count. No dynamical process in Phase Theory can reduce $N_{\text{adm}}[\Phi]$, because doing so would require creating a configuration with no admissible pre-image — a violation of the global topological invariance of the phase substrate. ■

13.2 Comparison with Other Approaches

Comparison with Hawking's original argument [6]. Hawking argued that the transition from a pure to a mixed density matrix during evaporation is unavoidable because the horizon separates the global quantum state into inaccessible parts. This argument is predicated on field-theoretic horizon physics: the partial trace over interior degrees of freedom of a quantum field. In Phase Theory, there is no partial trace over fields — there is one global phase configuration Φ . The apparent entropy increase of the exterior is a consequence of the limited external distinguishability, not of irreversible mixing.

Comparison with quantum mechanical unitarity. In quantum mechanics, unitarity is the statement that the time evolution operator U is unitary: $|\psi(t)\rangle = U(t)|\psi(0)\rangle$ with $U^\dagger U = 1$. In Phase Theory, unitarity is an effective principle valid when quantum fields approximate phase dynamics. The deeper conservation law is phase admissibility conservation (Theorem 13.4), not Hilbert-space unitarity. Phase admissibility conservation is more primitive than unitarity and reduces to it in the quantum effective description.

The Page curve. Information return from a black hole — the decrease of $S_{\text{rad}}(t)$ after the Page time — follows from phase equilibration dynamics in Phase Theory, as derived in Section 14. The Page time corresponds to the phase scrambling time (12.2), consistently.

14. The Page Curve from Phase Equilibration

14.1 Entropy of Emitted Radiation

Define $S_{\text{rad}}(t)$ as the entropy of emitted phase modes up to time t . This is the entropy that an external observer assigns to the cumulative output of phase leakage from the black hole. Phase Theory gives the following behavior:

Early regime ($t \ll t_{\text{Page}}$): $S_{\text{rad}}(t)$ increases approximately as:

$$S_{\text{rad}}(t) \approx N_{\text{modes}}(t) \cdot \log 2 \quad (\text{each new mode adds approximately one bit}) \quad (14.1)$$

because early modes carry essentially no correlations with interior configurations — they are drawn from near-boundary marginally admissible modes, which are topologically minimal and carry negligible interior information.

Late regime ($t \gg t_{\text{Page}}$): $S_{\text{rad}}(t)$ decreases as correlations between emitted modes and the shrinking interior become dominant in the leakage spectrum. The radiation is no longer approximately thermal — it encodes the interior topology progressively.

Proposition 13.6 (Page Curve from Phase Equilibration)

The Page curve follows from the competition between (i) growing radiation entropy from uncorrelated mode leakage in the early regime, and (ii) increasing phase correlation transfer from the interior as interior topology becomes accessible in the late regime. The crossover occurs at the Page time:

$$t_{\text{Page}} \approx G^2 M^3 / (\hbar c^4) \propto \tau_{\text{evap}} / 2 \quad (14.2)$$

14.2 The Phase Equilibration Mechanism

The mechanism is as follows. Early emission is drawn from near-boundary marginally admissible modes, which are topologically shallow — they carry little interior information, because they sample only the outermost layer of the saturation boundary. As evaporation progresses, the interior phase topology becomes progressively accessible: deeper phase modes (further from the boundary, carrying more interior correlation) leak out with increasing correlation content. The crossover between shallow and deep mode leakage corresponds to the Page time.

No replica geometry is required. No island surface is invoked. The Page curve is a direct consequence of phase equilibration dynamics: the gradual transition from boundary-mode to interior-mode leakage as ρ_Φ decreases from $\rho_{\max}$ toward zero.

14.3 The Island Surface as Effective Description

The island formula [10, 11] in gravity-coupled replica calculations locates an "island surface" in the interior whose inclusion in the entropy calculation reproduces the Page curve. In Phase Theory, the island surface is an effective geometric description of the phase correlation transfer boundary — the locus separating shallow-mode from deep-mode leakage at a given stage of evaporation. It is not a fundamental geometric structure; it is a description of a phase-substrate process in the language of effective geometry. The island "emerges" from phase correlation transfer dynamics, not from a fundamental topological feature of spacetime.

15. Why Radiation Appears Thermal — Observational Coarse-Graining

15.1 Three Sources of Apparent Thermality

The Planckian character of observed Hawking radiation (equation 11.2) is an approximation arising from three compounding factors:

(i) Minimal external distinguishability of emitted modes. Phase modes leaking from near the saturation boundary are topologically minimal — they have the simplest phase structure consistent with being externally distinguishable. Minimal-structure modes distribute uniformly over the energy spectrum, producing a thermal distribution. This is a Phase Theory version of the microcanonical argument for thermal distributions: the least-structured phase modes are the most numerous.

(ii) Inaccessibility of internal correlations. Full phase correlations between emitted modes and interior configurations are not externally distinguishable in the stationary saturation regime (Definition 13.3). Even though emitted modes are correlated with specific interior configurations, the external operator D_{ext} cannot resolve these correlations. External observations are therefore averages over many interior configurations, producing an effective thermal state.

(iii) Coarse-grained observation. Macroscopic detectors integrate over many phase modes, averaging out non-thermal correlations. Even if individual phase leakage events carry non-thermal information, the macroscopic signal is dominated by the coarse-grained average, which is thermal.

Proposition 13.7 (Apparent Thermality)

Hawking radiation appears exactly thermal to coarse-grained external observers, but is exactly non-thermal when full phase correlation information is included. The coarse-grained radiation state satisfies:

$$\rho_{\text{cg}} = \text{Tr}_{\text{interior}}[|\Phi\rangle\langle\Phi|] \approx \rho_{\text{thermal}}(T_{\text{H}}) \quad \text{with deviations of order } N_{\text{adm}}^{-1} \quad (15.1)$$

15.2 Detectability of Non-Thermal Correlations

Non-thermal correlations in Hawking radiation become statistically significant — detectable in principle — under two conditions: (a) the evaporation is nearly complete ($M \rightarrow M_{\text{P}}$), so that N_{adm} is no longer exponentially large; and (b) a full quantum-level measurement of all emitted modes from the beginning of evaporation is performed. For astrophysical black holes, neither condition is observationally accessible: their $N_{\text{adm}} \propto \exp(A/4\ell_{\text{P}}^2)$ is unimaginably large, and a full-mode measurement is far beyond any foreseeable technology.

For microscopic or primordial black holes — particularly those near the end of their evaporation lifetime — non-thermal effects may be measurable in the gamma-ray and cosmic-ray spectrum. Phase Theory makes specific predictions for the non-thermal spectral signature (Section 27, Prediction P13-1), which differ from both GR+QFT predictions and from LQG near-Planck-mass corrections.

16. Firewalls and Complementarity — Resolution via Global Phase Topology

16.1 The AMPS Firewall Paradox

The firewall paradox of Almheiri, Marolf, Polchinski, and Sully [9] (hereafter AMPS) demonstrated a tension between three assumptions widely regarded

as individually compelling: (A) unitarity of black hole evaporation; (B) the equivalence principle (an infalling observer experiences no local drama at the horizon); and (C) the effective validity of quantum field theory away from the Planck scale. AMPS argued that if (A) holds — Hawking radiation is exactly unitary — then the early and late radiation modes must be maximally entangled, leaving no entanglement available for the infalling observer's interior mode. By entanglement monogamy, this requires the interior mode to be in a high-energy state — a "firewall" — violating (B).

Numerous proposals were advanced to resolve the AMPS paradox: black hole complementarity [7], fuzzballs [8], ER=EPR [28], non-violent unitarization [13], among others. Each involves either rejecting one of (A), (B), or (C), or introducing new principles. Phase Theory resolves the paradox without any of these moves.

16.2 The Phase Theory Resolution

The resolution has three parts:

Part (i): No independent Hilbert spaces. The AMPS argument assumes that the interior and exterior quantum degrees of freedom span independent Hilbert spaces — so that entanglement of radiation with the interior is separately accountable from entanglement of radiation with other radiation. This assumption is valid when quantum fields are fundamental, because the field Hilbert space can be factorized over spatial regions. In Phase Theory, there is one global phase configuration Φ . There is no interior Hilbert space and no exterior Hilbert space — there is one phase substrate. The assumption of independent Hilbert spaces is invalid.

Part (ii): Entanglement monogamy is not violated. The monogamy argument assumes that early radiation mode R_E is maximally entangled with late radiation mode R_L , and also with interior mode B , leading to a contradiction. In Phase Theory, "early radiation mode," "late radiation mode,"

and "interior mode" are not independent degrees of freedom — they are all aspects of the global phase configuration Φ . The apparent monogamy violation does not arise because the putative independently-entangled objects are not independently-existing objects in Phase Theory.

Part (iii): Equivalence principle is maintained. The local phase configuration at Σ_H is fully admissible by Definition 13.1. An infalling observer's local measurements detect no deviation from the sub-saturated environment. High-energy modes at the horizon would require local admissibility violation at Σ_H , which is globally forbidden by PT-5.

Theorem 13.5 (No-Firewall Theorem)

In Phase Theory, infalling observers encounter no high-energy firewall at the phase horizon Σ_H . The equivalence principle — no local drama at the horizon — is maintained globally and is fully consistent with information conservation (Theorem 13.4).

Proof.

The phase horizon Σ_H is a change in external distinguishability (Definition 13.3), not a physical surface. The local phase configuration at Σ_H is admissible by Definition 13.1: $\rho_\Phi = \rho_{\max}$ with $\mathcal{A}[\Phi_{\text{sat}}] = 0$, which is the boundary of admissibility but not a violation of it. An infalling observer's local phase measurement at Σ_H detects no distinguishable deviation from the surrounding phase environment — the local phase topology is smooth, consistent with the equivalence principle. A firewall would require a local concentration of high-energy phase modes at Σ_H , which would violate local admissibility. By PT-5, global consistency enforcement propagates any local admissibility violation as a phase deformation — which would be observable far from the horizon, contradicting the stationarity of the

saturation configuration. Therefore, no firewall exists. █

16.3 Black Hole Complementarity — Made Unnecessary

Black hole complementarity (Susskind–Thorlacius–Uglum [7]) was proposed to reconcile external unitarity with internal regularity by positing that no single observer can verify both, and that the two descriptions are therefore mutually consistent despite apparent contradiction. Phase Theory makes complementarity unnecessary: there is no contradiction to resolve. Unitarity (in the effective sense — phase admissibility conservation) and internal regularity are simultaneously maintained in one globally consistent phase configuration. No restriction on observers, no complementary descriptions, and no new postulate is required.

17. Interior Structure — Phase Topology Beyond the Horizon

17.1 The Interior as Structured Phase Topology

The interior of a black hole — the region inside Σ_H — is, in Phase Theory, a highly structured phase topology, not an empty geometric interior. It is maximally constrained by global admissibility, dynamically stable at saturation, and non-singular. The conventional picture of an interior spacetime filled with matter converging on a singularity at $r = 0$ is replaced by a phase-saturated region of uniform maximum phase curvature density.

Proposition 13.8 (Interior Phase Topology)

The interior of a phase-saturated region Ω is a compact phase topology with

maximal phase curvature density $\rho_\Phi = \rho_{\max}$, globally admissible everywhere, with no distinguished center point.

The absence of a privileged center is crucial. In GR, the singularity at $r = 0$ in Schwarzschild coordinates is a distinguished point (or ring, in the Kerr case) toward which all matter converges. In Phase Theory, $\rho_\Phi = \rho_{\max}$ uniformly throughout Ω — there is no gradient pointing toward a singular center. The interior is topologically uniform in the saturation sense. The notion of " $r = 0$ " corresponds, in the effective geometric description, to the naive extrapolation of the Schwarzschild coordinate r through the horizon in a regime where the effective metric description has no validity.

Definition 13.5 (Phase Core)

The **phase core** is the maximally saturated interior region of a black hole, characterized by uniform $\rho_\Phi = \rho_{\max}$ and maximal topological complexity — the highest phase information density consistent with global admissibility. The phase core is compact, non-singular, and dynamically stable.

17.2 Comparison with Quantum Gravity Interior Proposals

(i) String theory fuzzballs. The fuzzball program [8] proposes that black holes are horizon-scale string configurations with no horizon and no interior — the horizon is replaced by a string microstate structure. In Phase Theory terms, the fuzzball description is an effective account of phase topological structure near the saturation boundary, valid in a specific regime where string degrees of freedom approximate phase modes. Phase Theory differs in maintaining a phase horizon (as a distinguishability boundary) while agreeing on the absence of a GR singularity.

(ii) LQG polymer quantization. In loop quantum gravity [16, 17], the interior singularity is replaced by a quantum geometry bounce — the volume reaches a minimum non-zero value at the quantum "Planck star" and then re-expands. This is conceptually similar to the phase core of Definition 13.5: a minimum-volume, maximum-curvature, non-singular interior configuration. The differences are foundational: LQG derives the bounce from discrete area spectra and polymer quantization of the connection; Phase Theory derives the bounded interior from the global admissibility constraint PT-2.

(iii) Causal dynamical triangulation. CDT constructs interior geometry from a statistical sum over Lorentzian triangulations, producing a quantum geometry that avoids classical singularities by construction. Phase Theory provides an alternative non-geometric origin of interior structure, without discretizing spacetime — the interior structure is a property of phase topology, not of quantized geometry.

18. The No-Singularity Theorem

Theorem 13.6 (No-Singularity Theorem)

Phase Theory admits no spacetime singularities. For any physical phase configuration $\Phi \in \mathcal{C}_{\text{adm}}$ (the space of globally admissible configurations), the phase curvature density $\rho_{\Phi}(x) \leq \rho_{\text{max}} < \infty$ for all x in the phase substrate. Therefore, no physical observable diverges.

Proof.

By Axiom PT-2, $\mathcal{A}[\Phi] \geq 0$ for all $\Phi \in \mathcal{C}_{\text{adm}}$. By Definition 13.1 and Lemma 13.1, the

maximum admissible phase curvature density is $\rho_{\max}$, achieved uniquely at saturation. For all sub-saturated configurations, $\rho_\phi < \rho_{\max}$. For saturated configurations, $\rho_\phi = \rho_{\max}$. Therefore, $\rho_\phi \leq \rho_{\max} < \infty$ on all of $\mathcal{C}_{\text{adm}}$. Since all physical observables are functionals of Φ and ρ_ϕ (by PT-1 and the emergence framework of Papers 4-9), and ρ_ϕ is bounded above, all physical observables are finite. No observable diverges. ■

Corollary 13.2

The Penrose–Hawking singularity theorems [1, 2] do not apply in Phase Theory. They assume (a) that spacetime is the fundamental arena (not emergent) and (b) that classical energy conditions hold throughout — both non-primitive in Phase Theory and inapplicable near saturation.

18.1 What Happens "At the Singularity"

In GR, the curvature scalars diverge at the Schwarzschild singularity ($r = 0$): the Kretschner scalar $K = R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta} = 48 G^2 M^2 / (c^4 r^6) \rightarrow \infty$ as $r \rightarrow 0$. In Phase Theory, this divergence is a consequence of using the effective Schwarzschild metric (7.1)–(7.2) in a regime where it has no valid extension — namely, inside the phase saturation boundary where phase topology has no faithful Riemannian approximation. The "singularity" is a geometric artifact of the approximation, not a feature of the underlying phase substrate. It is analogous to the apparent "infinity" in a divergent Taylor series approximation used outside its radius of convergence — the series diverges; the function it approximates does not.

The curvature density ρ_ϕ in the interior reaches $\rho_{\max}$ uniformly — it does not have the $1/r^6$ divergent radial profile of the GR Kretschner scalar. The radial dependence of the effective metric is itself an effective description that

breaks down deep inside the saturation region. The correct description of the interior is the phase core (Definition 13.5), not any extrapolation of the Schwarzschild metric.

18.2 Implications for Causality and Geodesic Completeness

In GR, singularities imply geodesic incompleteness: timelike and null geodesics reach the singularity in finite affine parameter, and they cannot be extended. In Phase Theory, there are no singularities and therefore no geodesic incompleteness in the phase-substrate description. Infalling phase trajectories reach the saturation center (the phase core) in finite phase-time and are not "destroyed." They continue to evolve as part of the saturated phase topology, eventually re-emerging as phase leakage modes during evaporation.

The effective GR geodesic incompleteness is reproduced approximately: infalling observers reach the phase core in finite proper time, consistent with the GR calculation. But they do not encounter a singularity — they enter the maximally saturated phase core and their information becomes internally encoded in the phase topology. This encoding is reversible during evaporation.

19. Gravitational Collapse and Phase Saturation Onset

19.1 Reinterpreting Collapse

Gravitational collapse is conventionally described as the infall of matter under its own gravity, compressing the matter to infinite density at a central singularity within a horizon. In Phase Theory, collapse is the accumulation of phase inertia density beyond the critical threshold Φ_c , triggering the phase

saturation transition. No matter implodes. No singularity forms. A new phase configuration — the saturated state — is established.

19.2 Phase Collapse Dynamics

Collapse proceeds through three phases, each with a distinct regime of physical behavior:

Phase I (Pre-collapse): $\rho_\Phi < \rho_c$. The phase configuration is sub-saturated. GR provides an excellent effective description of this regime. The collapse of a stellar-mass body follows the Oppenheimer-Snyder solution at leading order. The phase inertia density increases as matter accumulates. No qualitative deviation from GR is expected or predicted in this regime for conventional gravitational wave or electromagnetic observations.

Phase II (Saturation onset): $\rho_\Phi \rightarrow \rho_c$. Phase admissibility approaches its boundary — $\mathcal{A}_{\mu\nu}[\Phi] \rightarrow 0$. The effective metric develops the apparent horizon behavior: $g_{tt} \rightarrow 0$ at the Schwarzschild radius r_s as seen by external observers. In GR language, the apparent horizon forms. In Phase Theory, this is the onset of the phase saturation transition. The effective metric description is still valid in this regime but begins to receive near-saturation corrections (Section 7).

Phase III (Post-saturation): $\rho_\Phi \rightarrow \rho_{\max}$. Phase horizon Σ_H forms as the external distinguishability boundary. The interior transitions to the saturated phase core (Definition 13.5). External observers see the apparent horizon stabilize. Phase leakage begins (Section 10), initiating the evaporation process (Section 12).

Proposition 13.9 (Collapse as Phase Transition)

Gravitational collapse is a **phase transition** in the phase substrate from a sub-saturated to a saturated configuration, occurring at the critical phase inertia

density ρ_c . The transition is continuous (second-order in the Ehrenfest classification) — no discontinuity in ρ_ϕ , only in its first derivative $\partial\rho_\phi/\partial t$.

19.3 Collapse Timescale

The collapse timescale as seen by external observers is:

$$t_{\text{collapse}} \rightarrow \infty \quad (\text{external frame, due to phase-time dilation near saturation onset}) \quad (19.1)$$

$$t_{\text{collapse}} \approx r_s / c \quad (\text{infalling frame}) \quad (19.2)$$

Both results reproduce the GR time dilation: external observers see the collapsing body's surface asymptotically approach r_s but never cross it; infalling observers cross in finite proper time. These are consequences of the effective metric (7.1) near phase saturation onset, not of GR directly.

19.4 Angular Momentum in Collapse

Rotating collapse produces a phase configuration with non-zero topological winding number around the rotation axis — a **phase vortex**. The vortex structure is the Phase Theory origin of the Kerr metric (treated fully in Section 20). As collapse proceeds, angular momentum conservation in Phase Theory follows from the topological conservation of the winding number — a consequence of global admissibility enforcement on rotationally-symmetric phase perturbations.

20. Rotating Black Holes – Phase Vorticity and Kerr Correspondence

20.1 Phase Vorticity

Definition 13.6 (Phase Vorticity)

The **phase vorticity** Ω_Φ of a saturated region is the topological winding number of the phase configuration around the axis of symmetry:

$$\Omega_\Phi = (1/2\pi) \oint_C \nabla \Phi \cdot d\mathbf{l} \quad (20.1)$$

where C is a closed loop encircling the rotation axis. Ω_Φ is a topological invariant — conserved by global admissibility enforcement (PT-5) under all continuous phase deformations that do not cross a saturation boundary.

A rotating black hole corresponds to a phase-saturated configuration with non-zero vorticity $\Omega_\Phi = a/M$ (in units of G/c^2), where a is the standard Kerr spin parameter (with units of length: $a = J/Mc$, J = angular momentum).

Proposition 13.10 (Kerr Correspondence)

Phase-saturated configurations with vorticity Ω_Φ reproduce the Kerr metric as the leading-order effective metric in the sub-saturation limit. The effective line element at leading order is the Boyer-Lindquist form:

$$ds^2 = -(1 - r_s r / \Sigma) c^2 dt^2 - (2 r_s r a \sin^2 \theta / \Sigma) c dt d\varphi + (\Sigma / \Delta) dr^2 + \Sigma d\theta^2 + (r^2 + a^2 + r_s r a^2 \sin^2 \theta / \Sigma) \sin^2 \theta d\varphi^2 \quad (20.2)$$

where $\Sigma = r^2 + a^2 \cos^2 \theta$, $\Delta = r^2 - r_s r + a^2$, and $r_s = 2GM/c^2$. This is the Kerr metric, recovered from Phase Theory without invoking GR, in the sub-saturation limit.

20.2 Ergosphere and Frame Dragging

The ergosphere — the region between the outer horizon r_+ and the static limit surface $r_{\text{static}} = G M/c^2 + \sqrt{(G^2 M^2/c^4 - a^2 \cos^2 \theta)}$ — corresponds in Phase Theory to the region where vorticity effects dominate phase distinguishability. In this region, infalling phase modes are entrained by the rotating saturation topology — they cannot remain phase-stationary in the rotating frame. This is the phase-theoretic origin of the frame-dragging effect, here seen as topological entrainment of phase modes by the phase vortex.

20.3 The Penrose Process and Vortex-Mode Leakage

The Penrose process — the extraction of rotational energy from a Kerr black hole — corresponds in Phase Theory to the emission of vortex-mode leakage: phase modes that carry positive vorticity are emitted, reducing Ω_Φ (and hence the spin parameter a), while negative-vorticity modes return to the interior. The net effect is a reduction in angular momentum and rotational energy, consistent with the Penrose process energy extraction and the Blandford–Znajek mechanism for jet production.

20.4 Extremal Kerr and Topological Uniqueness

The extremal Kerr case ($a = M$ in units $G = c = 1$) corresponds to maximally vorticose phase saturation: the phase horizon contracts to a single-layer distinguishability boundary — an Ω_Φ -maximized saturated configuration. In Phase Theory, this is a topologically distinct class of saturated configuration, characterized by degenerate horizons (the inner and outer horizons coincide) and zero Hawking temperature ($T_H = 0$ at extremality, consistent with Proposition 13.3 at $a = M$). The Kerr bound $a \leq M$ is recovered from Phase Theory as the condition $\Omega_\Phi \leq \Omega_{\text{max}}$, where Ω_{max} is the maximum vorticity consistent with global admissibility for given Φ_{inertia} .

21. Charged Black Holes — Phase Polarity Saturation

21.1 Charge as Phase Polarity

From Paper 7 (Gauge Interactions), electric charge is an emergent property of phase polarity — a topological winding number associated with the U(1) phase symmetry of the phase substrate. This is consistent with the Phase Theory derivation of electromagnetism in Paper 7, where the U(1) gauge field emerges from phase polarity gradients.

Definition 13.7 (Phase Polarity)

The **phase polarity** P_Φ of a phase configuration is its net topological U(1) winding number, corresponding to emergent electric charge:

$$Q = e \cdot P_\Phi \quad (\text{Paper 7, equation 7.14}) \quad (21.1)$$

where e is the fundamental charge unit, determined from the U(1) winding quantum of the phase substrate. P_Φ is conserved under all continuous phase evolutions by global admissibility enforcement (PT-5).

Proposition 13.11 (Reissner–Nordström Correspondence)

Phase-saturated configurations with polarity $P_\Phi \neq 0$ reproduce the Reissner–Nordström metric at leading order:

$$f(r) = 1 - r_s/r + r_Q^2/r^2 \quad (21.2)$$

where $r_Q^2 = GQ^2 / (4\pi\epsilon_0 c^4)$ follows from phase polarity accumulation in the effective description. This is the Reissner–Nordström metric function, recovered from Phase Theory.

21.2 Extremal Reissner-Nordström and Perfect Admissibility Balance

The extremal Reissner-Nordström case ($Q = M$ in Planck units, i.e., $r_Q = r_s/2$) corresponds to exact cancellation of phase inertia and polarity contributions at the saturation boundary. In Phase Theory, this corresponds to a configuration where the outward phase inertia pressure (driving leakage) is exactly balanced by the inward phase polarity confinement (suppressing leakage of charged modes). The result is $T_H = 0$ — Hawking radiation ceases for the extremal charged black hole, consistent with equation (10.2) at extremality (which gives $\kappa = 0 \rightarrow T_H = 0$ by Proposition 13.3).

21.3 The Kerr-Newman Family and Topological Parameter Space

The Kerr-Newman family (rotating, charged) corresponds to simultaneous nonzero vorticity Ω_Φ and polarity P_Φ — a doubly topological phase-saturated configuration. The effective metric is the Kerr-Newman solution (a generalization of equations 20.2 and 21.2 to include both parameters), reproduced from Phase Theory in the sub-saturation limit.

The full parameter space $\{M, a, Q\}$ of the Kerr-Newman family is encoded in Phase Theory as $\{\Phi_{\text{inertia}}, \Omega_\Phi, P_\Phi\}$ — the three topological invariants of a macroscopic phase-saturated configuration. This correspondence is not coincidental: it is the Phase Theory derivation of the no-hair theorem, established rigorously in Section 22.

22. No-Hair Theorem from Phase Topology

Theorem 13.7 (Phase No-Hair Theorem)

A macroscopically phase-saturated configuration is uniquely characterized externally by three topological invariants: $\{\Phi_{\text{inertia}}, \Omega_\Phi, P_\Phi\}$ — corresponding to mass M , angular momentum J ($= aM$), and charge Q . No additional external structure ("hair") is possible.

Proof.

By global admissibility enforcement (PT-5), all interior admissible configurations consistent with boundary constraints $\{\Phi_{\text{inertia}}, \Omega_\Phi, P_\Phi\}$ are equally weighted in the phase entropy count (Section 8). The external distinguishability boundary Σ_H encodes exactly the three topological invariants $\{\Phi_{\text{inertia}}, \Omega_\Phi, P_\Phi\}$: Φ_{inertia} determines Σ_H 's total area (via Theorem 13.2), Ω_Φ determines its vortex structure (Definition 13.6), and P_Φ determines its polarity structure (Definition 13.7). Any higher multipole phase structure in the interior maps to the same exterior phase state by Definition 13.3 — the external operator D_{ext} is degenerate on higher multipoles. Therefore, the exterior is entirely determined by $\{\Phi_{\text{inertia}}, \Omega_\Phi, P_\Phi\}$. ■

Corollary 13.3

"Hair" — additional external structure beyond $\{M, J, Q\}$ — is forbidden in Phase Theory because additional interior topology is externally indistinguishable (Proposition 13.1 and Definition 13.3). Black holes are topologically bald.

22.1 Hair Violations in Effective Field Theory

In effective field theories, scalar hair and Yang-Mills hair can exist — specific field configurations can produce observable external structure beyond $\{M, J,$

Q}. These are consistent with Phase Theory: scalar hair corresponds to a non-zero expectation value of an emergent scalar field (a phase mode with scalar-like topology) outside the black hole, and Yang-Mills hair corresponds to non-Abelian phase polarity configurations. In Phase Theory, all such field-theoretic hair is ultimately reducible to the three topological invariants at the fundamental level — "hair" in effective field theory corresponds to metastable higher-multipole phase structures that are externally distinguishable in the presence of long-range field condensates. This is a perturbative effect, suppressed by the hair-mode mass and relevant only at distances far from the saturation boundary.

23. Comparison with General Relativity

23.1 Agreements

Phase Theory reproduces all classical GR results for black holes as effective descriptions in the sub-saturation, low-curvature regime:

- Schwarzschild, Kerr, and Reissner-Nordström metrics (Sections 7, 20, 21)
- Bekenstein-Hawking entropy $S_{\text{BH}} = A/4\ell_{\text{P}}^2$ (Theorem 13.2)
- Hawking temperature $T_{\text{H}} = \hbar\kappa / (2\pi k_{\text{B}})$ (Proposition 13.3)
- Orbital dynamics: innermost stable circular orbit $r_{\text{ISCO}} = 3r_{\text{s}}$ (effective metric calculation)
- Photon sphere: $r_{\text{ph}} = 3GM/c^2$ at leading order (effective metric calculation)
- Gravitational redshift: $z + 1 = (1 - r_{\text{s}}/r)^{-1/2}$ (effective metric calculation)
- Black hole area theorem: A non-decreasing in classical processes (follows from S_{BH} non-decreasing)

- Gravitational wave ringdown frequencies from quasi-normal modes (effective metric)

23.2 Disagreements and Extensions

Aspect	General Relativity	Phase Theory
Singularities	Inevitable by Penrose–Hawking theorems; infinite curvature	Forbidden by global admissibility (Theorem 13.6); $\rho_\Phi \leq \rho_{\max}$
Information	Lost in evaporation (Hawking 1976); or requires non-local mechanisms	Conserved by phase topology (Theorem 13.4); recovered during evaporation
Horizon status	Geometric artifact in some coordinates; thermodynamic surface in others	Phase distinguishability boundary (Definition 13.3); no physical surface
Near-saturation metric	No modification; GR valid all the way to singularity	Corrections of order $(\rho_\Phi/\rho_{\max})$ near horizon (equation 7.4)
Evaporation endpoint	Remnant or information loss	Complete evaporation with information recovery (Theorem 13.3)
Interior	Spacetime region with converging matter, singular endpoint	Phase core: uniform maximal saturation, no singularity, no center

23.3 On the Penrose Singularity Theorem

Penrose's 1965 theorem [1] demonstrates that, given: (a) a spacetime satisfying the Einstein field equations, (b) the null energy condition $R_{\mu\nu}k^\mu k^\nu \geq 0$, and (c) the existence of a non-compact Cauchy surface, the presence of a trapped surface implies a singularity. Phase Theory does not violate this theorem — it renders it inapplicable. Premise (a) fails: Phase Theory does not posit the Einstein field equations as fundamental. Premise (b) fails: near

saturation, the effective null energy condition receives corrections from the phase saturation tensor $P_{\mu\nu}$ (equation 4.3) that can violate the NEC. The theorem's premises do not hold in Phase Theory; its conclusion is therefore not derived.

24. Comparison with String Theory and Holography

24.1 Fuzzball Program

Mathur's fuzzball program [8] proposes that each quantum microstate of a black hole corresponds to a distinct string configuration — a "fuzzball" — that fills the would-be horizon volume with string matter, leaving no empty horizon or interior. Phase Theory agrees with the no-singularity conclusion, but differs structurally: Phase Theory maintains a phase horizon as a distinguishability boundary (Theorem 13.1), whereas the fuzzball proposal removes the horizon entirely (replacing it with a stringy surface). The two frameworks may be compatible at the effective level: fuzzballs are plausibly an effective description of phase topological structure near the saturation boundary in a string-theoretic approximation regime.

24.2 AdS/CFT Correspondence

The AdS/CFT correspondence [22] has been used to study black holes holographically: a large AdS black hole is dual to a thermal state in the boundary CFT; its Hawking temperature corresponds to the CFT temperature; its entropy corresponds to CFT thermal entropy. Phase Theory agrees with these correspondences at the thermodynamic level but derives them from phase saturation rather than string duality. Phase Theory holography (Corollary 13.1) is universal — it does not require AdS asymptotics, specific gauge groups, or large-N limits.

24.3 Replica Wormholes and Islands

Recent calculations [10, 11] show that the Page curve can be reproduced using replica geometry and island surfaces in JT gravity coupled to a CFT. The key step is the inclusion of Euclidean wormholes (replica wormholes) connecting different replicas in the path integral, which effectively computes the entropy of the radiation including island contributions. Phase Theory derives the same Page curve from phase equilibration dynamics (Section 14) without replica geometry or island surfaces — these are effective geometric descriptions of phase correlation transfer processes. The island formula is a geometric shadow of the phase correlation leakage, not a fundamental geometric fact.

24.4 ER=EPR Conjecture

The Maldacena-Susskind ER=EPR conjecture [28] proposes that quantum entanglement between two systems is geometrically realized as an Einstein-Rosen bridge connecting them. In Phase Theory, quantum entanglement is an effective description of phase correlation between non-adjacent (in the effective geometric sense) phase regions. ER bridges correspond to topologically connected phase configurations: two phase-saturated regions sharing a topological connection in the phase substrate correspond, in the effective geometric description, to an Einstein-Rosen bridge. ER=EPR is therefore a consequence of Phase Theory, not an independent conjecture.

25. Comparison with Loop Quantum Gravity

25.1 LQG Black Holes

Loop quantum gravity [16, 17] provides one of the most developed quantum gravity treatments of black holes. The LQG approach discretizes area and

volume via spin networks and spin foams, replacing the classical continuum geometry with a discrete quantum structure. The singularity is addressed via polymer quantization of the interior, which replaces the classical singularity with a quantum bounce — the "Planck star" proposal [18].

25.2 Comparison

Aspect	Loop Quantum Gravity	Phase Theory
Singularity resolution	Quantum bounce from polymer quantization; Planck star	Phase core from global admissibility; no bounce required
Area spectrum	Discrete: $A_n = 8\pi\gamma\ell_P^2 \sqrt{j(j+1)}$; γ = Barbero-Immirzi parameter	Discrete phase mode spectrum from phase quantization (Paper 2)
Information paradox	Commitment to unitarity; incomplete resolution	Complete structural resolution via Theorem 13.4
Entropy	$S = A/(4G\hbar) + \log$ corrections; γ fixed by matching to BH entropy	$S = A/(4\ell_\phi^2) - (3/2)\log(A/\ell_\phi^2)$; coefficient $-3/2$ predicted
Hawking spectrum	Non-thermal corrections from discrete area spectrum	Non-thermal corrections from phase topology correlations
Spacetime	Discrete quantum geometry; spin network/foam	Emergent from phase substrate; no quantized geometry

25.3 Logarithmic Entropy Corrections

Both LQG and Phase Theory predict logarithmic corrections to the Bekenstein-Hawking entropy, but with different coefficients. LQG typically gives [17]:

$$S_{\text{LQG}} = A/4\ell_P^2 - (1/2) \log(A/\ell_P^2) + O(\ell_P^2/A) \quad (25.1)$$

Phase Theory predicts (equation 8.7):

$$S_{\text{PT}} = A/4\ell_{\Phi}^2 - (3/2) \log(A/\ell_{\Phi}^2) + O(\ell_{\Phi}^2/A) \quad (25.2)$$

The differing logarithmic coefficients ($-1/2$ vs. $-3/2$) provide a potential discriminating test between the frameworks, if the logarithmic term in black hole entropy can be measured independently of the area term — for example, through analogue black hole experiments (Prediction P13-5).

26. Comparison with Causal Set Theory and Other Approaches

26.1 Causal Set Theory

Causal set theory (Bombelli, Lee, Myrheim, Sorkin, 1987 [29]) replaces the spacetime manifold with a discrete partial order — a causal set — in which each element corresponds to a spacetime event and the partial order encodes causal precedence. Like Phase Theory, causal set theory treats spacetime as an approximation to a more fundamental discrete structure. Key differences: (a) causal set theory retains causal order as a fundamental primitive; Phase Theory derives causal structure from phase topology (Paper 5). (b) Causal sets are discrete; the phase substrate has a different kind of discreteness (phase mode quantization) that does not directly correspond to a discrete point set. (c) Entropy in causal sets is linked to the number of order-preserving embeddings; in Phase Theory, entropy is the count of admissible phase configurations.

26.2 Causal Dynamical Triangulation

Causal dynamical triangulation (CDT) constructs a non-perturbative quantum gravity path integral from a sum over Lorentzian triangulations, with a preferred arrow of time imposed on each simplex. The resulting geometry is

emergent from the sum. Phase Theory is more radical in its emergence program: geometry is entirely emergent from phase structure, and the phase substrate itself is not geometric. CDT retains the geometric character of spacetime as the language of the path integral; Phase Theory abandons geometry at the fundamental level.

26.3 Non-Commutative Geometry

Connes' non-commutative geometry [30] proposes that spacetime is described by a non-commutative algebra, with the usual manifold recovered as a commutative limit. In Phase Theory, phase substrate coordinates satisfy uncertainty-like relations (Paper 2, Theorem 2.7), which produce effective non-commutativity at the admissibility scale Λ_Φ . Non-commutativity is therefore a consequence of Phase Theory, not a premise. The two frameworks are potentially compatible in a limit where phase topology reduces to an algebraic structure.

26.4 Asymptotic Safety

The asymptotic safety program (Reuter, 1998 [31]) proposes that gravity is UV-complete via a non-Gaussian fixed point of the gravitational renormalization group. The fixed point provides a maximal effective Newton constant $G_{UV} < \infty$ at high energies, regularizing otherwise divergent curvatures. Phase Theory avoids this entirely by rejecting fields — including the gravitational field — as fundamental. There is no renormalization group in Phase Theory because there is no field theory to renormalize. The effective GR is non-renormalizable in field-theoretic terms, but Phase Theory provides a UV completion without field-theoretic regularization.

27. Observable Consequences and Falsifiable Predictions

We enumerate eight classes of Phase Theory predictions for black holes with current or near-future observational relevance. Each prediction is stated precisely, its Phase Theory derivation is cited, and the discriminating comparison with GR+QFT is noted.

Prediction P13-1: Non-Thermal Correlations in Hawking Radiation

Late-stage radiation from microscopic and primordial black holes exhibits non-Planckian spectral correlations. The correlation signature follows the phase equilibration pattern of Section 14: systematic non-thermal excess at frequencies $\omega > k_B T_H / \hbar$, with a correlation spectrum scaling as $N_{\text{adm}}^{-1/2}$. This differs from the thermal spectrum of GR+QFT and from LQG discrete-area corrections. Observable in principle from primordial black hole evaporation events.

Prediction P13-2: Absence of GR Singularity Signatures in Ringdown

Post-merger gravitational wave ringdown signals should show quasi-normal mode overtone structure consistent with a non-singular, phase-saturated interior rather than a GR singularity. Specifically, the overtone damping times should deviate from GR predictions at the level of $(\ell_\Phi/r_s)^2$ in the quasi-normal mode imaginary part. This is sub-Planckian for stellar-mass black holes but potentially resolvable with next-generation detectors.

Prediction P13-3: Saturation Corrections to Photon Sphere

Phase Theory predicts a small deviation from the GR photon sphere radius:

$$\delta r_{\text{ph}} \approx \alpha (\ell_{\Phi}/r_s)^2 r_s \quad (27.1)$$

For stellar-mass black holes, this correction is $\sim (10^{-35}/10^3)^2 \times 10^3 \text{ m}$ — unmeasurably small. For primordial black holes near Planck mass, it approaches $O(1)$ relative correction.

Prediction P13-4: Phase Horizon Gravitational Wave Echoes

Gravitational wave echoes following post-merger ringdown arise from reflections of gravitational wave modes at the phase saturation boundary Σ_H . The echo time delay is:

$$\Delta t \approx (r_s / c) \cdot \log(r_s / \ell_{\Phi}) \approx 2GM/c^3 \cdot \log(M/M_P) \quad (27.2)$$

For a $10 M_{\odot}$ black hole, $\Delta t \approx 10^{-4} \text{ s} \times \log(10^{66}) \approx 3 \times 10^{-2} \text{ s}$. Echo amplitude is suppressed by the phase transmission coefficient $\tau(\omega)$.

Prediction P13-5: Modified Entropy-Area Relation

The sub-leading logarithmic correction to Bekenstein-Hawking entropy has coefficient $-3/2$ in Phase Theory (equation 8.7), distinct from the $-1/2$ of certain LQG predictions and the exact string-theoretic results for extremal black holes. Measurement via analogue systems could discriminate.

Prediction P13-6: Information Recovery Spectrum After Page Time

After $t > t_{\text{Page}}$, Hawking radiation shows increasing non-thermal correlations following a specific Phase Theory pattern: the correlation function $C(\omega_1, \omega_2)$ between modes at frequencies ω_1 and ω_2 grows as $(t/t_{\text{Page}} - 1)^{1/2}$ at leading order.

This functional form differs from GR+QFT predictions and provides a direct observational test of the phase equilibration mechanism.

Prediction P13-7: Absence of Firewall Signatures for Infalling Observers

No high-energy particle flux at the horizon is predicted for infalling observers (Theorem 13.5). In analogue gravity experiments with appropriate quantum systems, probes crossing the analogue horizon should detect no excess energy density. Phase Theory predicts exact equivalence principle satisfaction at Σ_H .

Prediction P13-8: Phase Vorticity Signature in Kerr Evaporation

Rotating black holes evaporate angular momentum and mass. The ratio $dJ/dt \div dM/dt$ in Phase Theory is:

$$(dJ/dt) / (dM/dt) = a \cdot (1 + \beta (a/M)^2 (\ell_\phi/r_s)^2) \quad (27.3)$$

where β is a Phase Theory parameter of order unity. The correction at sub-leading order in $(\ell_\phi/r_s)^2$ differs from the GR+QFT result by the β term, in principle measurable for rapidly rotating stellar-mass black holes over astrophysical timescales.

28. Experimental Protocols

For each prediction class of Section 27, we specify the experimental protocol, the required sensitivity, and the primary observational platform.

Protocol P13-1: Primordial Black Hole Spectroscopy

Monitor gamma-ray and cosmic-ray spectra for signatures of primordial black hole evaporation. The near-terminal spectrum (last seconds of evaporation for $M \sim 10^{12}$ kg PBHs completing evaporation today) should show systematic non-Planckian excess. Instruments: *Fermi* LAT, Cherenkov Telescope Array (CTA), HAWC observatory. Discriminator: spectral shape deviation from Planck distribution at late-time signal; look for systematic correlation excess at $\omega > 3k_B T_H/\hbar$ inconsistent with greybody-modified Planck spectrum.

Protocol P13-2: LIGO/ET Quasi-Normal Mode Analysis

Analyze post-merger gravitational wave signals from binary black hole coalescences for overtone structure. Compare best-fit quasi-normal mode frequencies and damping times with GR predictions and Phase Theory predictions including saturation corrections. Required: SNR > 100 for overtone resolution, achievable with Einstein Telescope (ET) or Cosmic Explorer (CE). Focus on events with remnant mass $10\text{--}50 M_\odot$ where ratio r_s/ℓ_Φ is maximally constraining.

Protocol P13-3: Event Horizon Telescope High-Resolution Imaging

Use EHT and next-generation EHT (ngEHT) imaging of M87* ($M \approx 6.5 \times 10^9 M_\odot$) and Sgr A* ($M \approx 4.1 \times 10^6 M_\odot$) to constrain photon ring radius to sub-percent precision. Phase Theory correction $\delta r_{\text{ph}}/r_{\text{ph}} \sim (\ell_\Phi/r_s)^2$ is unmeasurably small for these objects, but improvements in imaging resolution may constrain the phase-saturation correction parameter α at the level of Phase Theory predictions.

Protocol P13-4: Gravitational Wave Echo Search

Search for gravitational wave echoes in LIGO/Virgo/KAGRA data following binary black hole mergers. Predicted echo time delay: $\Delta t \approx (2GM/c^3) \log(M/M_P)$ (equation 27.2). For typical mergers with $M_{\text{rem}} \sim 30 M_\odot$, $\Delta t \approx 0.1$ s.

Search method: matched-filter analysis with Phase Theory echo templates in the post-merger data window Δt to $\Delta t + r_s/c$. Compare with GR+QFT (no echo) and other quantum gravity echo predictions.

Protocol P13-5: Analogue Black Hole Entropy Measurement

In tabletop analogue black hole experiments (Steinhauer acoustic black holes [32] or photonic analogue systems), measure entropy of Hawking-analogue radiation as a function of analogue horizon area A_{analogue} . Fit $S(A_{\text{analogue}})$ to the functional form $C_1 A_{\text{analogue}} + C_2 \log A_{\text{analogue}} + C_3$. Phase Theory predicts $C_2/C_1 = -(3/2) \ell_{\text{analogue}}^2$ (with appropriate analogue Planck length). Compare C_2 with LQG ($-1/2$) and string theory (varies) predictions.

Protocol P13-6: Full-Mode Tomography in Analogue Systems

Develop quantum information theory protocols for full-mode tomography of Hawking-analogue radiation in condensed matter analogue systems. Measure the two-mode correlation function $C(\omega_1, \omega_2)$ between emitted modes as a function of time (analogue evaporation progression). Look for the transition from uncorrelated (thermal) to correlated (non-thermal) emission consistent with the Phase Theory prediction of growing correlations after the analogue Page time. This protocol requires state-of-the-art quantum sensing and mode-resolution technology.

Protocol P13-7: Analogue Firewall Probe

In analogue gravity experiments with appropriate quantum systems (Bose-Einstein condensate analogues or optical analogues), prepare an infalling probe degree of freedom and measure its energy distribution as it traverses the analogue horizon. Phase Theory predicts no excess energy density at the analogue Σ_H . Any significant excess would be inconsistent with Phase Theory and potentially consistent with AMPS firewall predictions. Discriminating

between Phase Theory and other no-firewall proposals (e.g., complementarity) requires mode-specific measurements beyond analogue horizon traversal statistics.

Protocol P13-8: Kerr Spin-Down Measurement

Track spin-down rate of rapidly rotating astrophysical black holes in X-ray binaries via Fe $K\alpha$ line reverberation mapping and continuum spectroscopy. The spin parameter a/M as a function of time constrains dJ/dt at fixed mass. Phase Theory predicts the correction factor β in equation (27.3) is of order unity, producing a sub-leading deviation in the ratio $(dJ/dt)/(dM/dt)$ relative to GR+QFT. This requires multi-decade monitoring programs with X-ray telescopes at sensitivity levels beyond *RXTE* and achievable with *STROBE-X* or *eXTP*.

29. Implications for Cosmology and the Early Universe

29.1 Primordial Black Holes as Phase Relics

Primordial black holes (PBHs) formed in the early universe are, in Phase Theory, phase-saturated regions seeded by over-dense phase inertia fluctuations in the primordial phase substrate. They form when local phase curvature density fluctuations $\delta\rho_\phi$ exceed the critical saturation threshold at horizon crossing during the radiation-dominated era. The PBH mass function — the distribution of initial PBH masses — is determined by the power spectrum of phase inertia fluctuations, which in Phase Theory is related to (but not identical with) the curvature power spectrum $P_\zeta(k)$ of inflationary cosmology (Paper 10).

Phase Theory predicts that PBH formation is slightly more efficient than standard GR calculations suggest, because the phase saturation transition is a second-order (continuous) phase transition rather than the sharp gravitational collapse assumed in GR. The critical collapse threshold δ_c in Phase Theory is modified by phase saturation corrections of order $(\ell_\Phi/r_H)^2$, where r_H is the Hubble radius at formation. For practical PBH formation calculations in the Standard Model era, this correction is negligible; it becomes relevant only for PBHs forming near the Planck epoch.

29.2 PBHs as Dark Matter and Phase Theory Constraints

If a significant fraction of dark matter consists of primordial black holes in the mass window 10^{17} – 10^{22} kg (the asteroid-mass window, currently observationally allowed [33]), Phase Theory predicts modified evaporation histories compared to GR+QFT: specifically, the late-stage evaporation spectrum includes the non-thermal Phase Theory corrections of Prediction P13-1. The modified dark matter power spectrum imprints on small-scale structure at comoving scales corresponding to PBH formation epochs. Next-generation 21-cm cosmology experiments (SKA, HERA) can constrain the PBH mass function and potentially detect Phase Theory spectral signatures.

29.3 The Big Bang as Global Phase De-Saturation

A speculative but structurally motivated extension of Phase Theory is the following: the Big Bang itself may correspond to a global phase de-saturation event — a transition from a uniform, maximally phase-saturated initial state of the phase substrate to a differentiated, sub-saturated configuration. Under this interpretation, the initial singularity of standard cosmology is replaced by a globally saturated phase state ($\rho_\Phi = \rho_{\max}$ everywhere) — the cosmological analogue of the black hole phase core. The Big Bang corresponds to the onset of phase leakage from this initial saturated state, producing the hot, dense, but non-singular beginning of the observable universe. This hypothesis will be

developed in full in Paper 14 (Cosmological Phase Saturation and the Big Bang).

29.4 Black Hole Mergers and Phase Topology

When two black holes merge, their phase topologies combine. The pre-merger pair corresponds to two distinct phase-saturated regions $\{\Omega_1, \Omega_2\}$ with topological invariants $\{M_1, J_1, Q_1\}$ and $\{M_2, J_2, Q_2\}$. The merger produces a combined saturated region Ω_f with invariants determined by conservation of total phase inertia, vorticity, and polarity. The ringdown gravitational wave signal encodes the relaxation of the combined phase topology to the new stable Kerr saturation configuration. Phase Theory predictions for ringdown quasi-normal modes and their overtones (Protocol P13-2) are testable with next-generation detectors.

29.5 Hawking Radiation and the Baryon Asymmetry

A speculative but structurally motivated observation: phase leakage from very early, very small black holes may have seeded the baryon asymmetry if phase vorticity preferentially leaks in one topological direction — i.e., if there is a phase-theoretic CP violation associated with vorticity emission. In Phase Theory, CP symmetry is a consequence of the combined charge conjugation (phase polarity reversal $P_\Phi \rightarrow -P_\Phi$) and parity (spatial winding reversal) symmetry of the phase substrate. If the initial phase vortex distribution in the early universe was not CP-symmetric — e.g., due to a non-zero global phase chirality inherited from the initial saturated state — then vorticity leakage from early PBHs would produce a net baryon asymmetry. This is highly speculative and deferred to future work in the Phase Theory series.

30. Connections to the Phase Theory Series — Papers 1-12

This paper is Paper 13 in the Phase Theory series. We explicitly situate its results within the cumulative framework established by Papers 1-12, and preview Papers 14-15.

Paper	Title (Hanks, 2026)	Role in This Paper
Paper 1	Phase Primacy	Axioms PT-1 through PT-5; foundational premise throughout
Paper 2	Phase Topology	Topological invariants; basis for horizon distinguishability (Section 6) and no-singularity theorem (Section 18)
Paper 3	Emergence of Spacetime	GR as effective description; black holes as geometric approximations of phase-saturated configurations (Sections 7, 23)
Paper 4	Quantum Mechanics from Phase	Unitarity as effective principle; Phase Theory resolution of information loss without Hilbert-space arguments (Section 13)
Paper 5	Special Relativity from Phase	Causal structure of horizons from phase distinguishability constraints (Section 6)
Paper 6	Particle Spectrum	Hawking radiation modes as phase relaxation modes; mode spectrum (Section 10)

Paper	Title (Hanks, 2026)	Role in This Paper
Paper 7	Gauge Interactions	Charge as phase polarity; charged black hole analysis (Section 21)
Paper 8	Phase Thermodynamics	Entropy and temperature definitions; Bekenstein-Hawking derivation (Sections 8–11)
Paper 9	Gravitational Dynamics	GR recovery; metric corrections; underpins Sections 7, 12, 22
Paper 10	Phase Cosmology	Early universe context for PBH formation (Section 29)
Paper 11	Dark Sector	PBH dark matter connections; dark matter power spectrum (Section 29.2)
Paper 12	Quantum Gravity Foundations	Phase quantization; technical apparatus for Sections 5, 17, 18; Lemma 12.1, Theorem 12.4, 12.6, 12.9

Future papers using results of this paper:

- Paper 14** (Cosmological Phase Saturation and the Big Bang): Uses the phase de-saturation framework of Section 29.3 and the results of Sections 3, 18, and 19 to reformulate the Big Bang singularity as a cosmological phase saturation onset.
- Paper 15** (Phase Theory of Neutron Stars and Compact Objects): Uses the sub-saturation phase dynamics of Sections 4 and 19 to describe neutron stars as near-critical ($\rho_\Phi \rightarrow \rho_c$) phase configurations, with implications for the neutron star equation of state and gravitational wave signatures of neutron star mergers.

31. Summary of Results

We enumerate all principal results derived in this paper:

1. **Phase saturation replaces singularities** (Theorem 13.6, Section 18): $\rho_\Phi \leq \rho_{\max} < \infty$ everywhere; no physical observable diverges in Phase Theory.
2. **Phase horizons are distinguishability boundaries, not physical surfaces** (Definition 13.3, Theorem 13.1, Section 6): Σ_H is defined by the locus where $\mathcal{A}_{\mu\nu}[\Phi] \rightarrow 0$; the radial phase current vanishes there; no infalling observer detects a physical surface.
3. **Black hole entropy reproduces the Bekenstein-Hawking formula from phase configuration counting** (Theorem 13.2, Section 8): $S_{\text{BH}} = A/4\ell_\Phi^2$ with logarithmic correction $-(3/2) \log(A/\ell_\Phi^2)$; derived purely from phase boundary degrees of freedom.
4. **Holographic principle is a structural theorem in Phase Theory** (Corollary 13.1, Section 9): holography follows from distinguishability collapse at phase horizons; not an independent postulate.
5. **Hawking radiation is phase leakage, not pair creation** (Definition 13.4, Proposition 13.3, Section 10): no vacuum fluctuations, no particle creation; radiation is admissibility relaxation at Σ_H driven by global consistency enforcement.
6. **Hawking spectrum is Planckian at leading order with non-thermal corrections** (Proposition 13.4, Section 11): $dN/d\omega \propto \omega^2 [\exp(\hbar\omega/k_B T_H) - 1]^{-1} \times g(\omega)$; corrections suppressed by N_{adm}^{-1} .
7. **Evaporation proceeds without remnant** (Theorem 13.3, Section 12): ρ_Φ decreases continuously to zero; no topological obstruction at the end of evaporation; no stable remnant.

8. **Information is never destroyed** (Theorem 13.4, Section 13): $N_{\text{adm}}[\Phi]$ is a topological invariant of the phase substrate; conserved under all admissible dynamics; information becomes accessible during evaporation.
 9. **Page curve reproduced from phase equilibration without replica geometry** (Proposition 13.6, Section 14): competition between uncorrelated early leakage and correlated late leakage reproduces $S_{\text{rad}}(t)$ with maximum at $t_{\text{Page}} \approx \tau_{\text{evap}}/2$.
 10. **Firewalls do not arise** (Theorem 13.5, Section 16): interior and exterior are not independent Hilbert spaces; entanglement monogamy violation does not occur; equivalence principle maintained globally.
 11. **Interior is structured, non-singular phase topology** (Proposition 13.8, Definition 13.5, Section 17): phase core has uniform $\rho_\Phi = \rho_{\text{max}}$, maximal topological complexity, no center, no singularity.
 12. **No-hair theorem follows from phase topological invariants** (Theorem 13.7, Corollary 13.3, Section 22): macroscopic saturated configurations are characterized uniquely by $\{\Phi_{\text{inertia}}, \Omega_\Phi, P_\Phi\} \leftrightarrow \{M, J, Q\}$.
 13. **Kerr and Reissner-Nordström families recovered from phase vorticity and polarity** (Propositions 13.10, 13.11, Sections 20–21): rotating and charged black holes are doubly-topological saturated configurations; GR metrics reproduced at leading order.
 14. **Eight classes of falsifiable predictions are enumerated with experimental protocols** (Sections 27–28): predictions P13-1 through P13-8 with specific functional forms, required sensitivities, and instrument specifications.
-

32. Conclusion

This paper has reformulated black holes entirely within Phase Theory as macroscopically stable, globally admissible, phase-saturated configurations governed by the global consistency constraint $\mathcal{A}[\Phi] \geq 0$. Every major paradox of black hole physics has been shown to arise from importing non-primitive constructs — spacetime, quantum fields, vacuum fluctuations — into regimes where phase saturation dominates and those constructs fail. Phase Theory provides a unified, non-singular, information-preserving, firewall-free description of black holes that recovers GR as an effective shadow and generates novel, falsifiable predictions beyond GR and QFT.

The results are summarized starkly: Black holes do not destroy information. They do not harbor singularities. Their horizons are not surfaces. They are maximal phase configurations — the most structured, most constrained, most admissible states accessible to the phase substrate. Their eventual evaporation is the thermodynamically inevitable redistribution of phase curvature toward equilibrium, with full information recovery, no catastrophe, and no loss. The seeming mysteries of black hole physics — the impossibly dense interior, the one-way membrane, the lost information, the fiery horizon — are artifacts of applying approximate descriptions beyond their domain of validity.

Phase Theory does not patch GR. It does not regularize a divergent field theory. It does not add a quantum correction to a classical background. It replaces the ontological foundation: Φ is the sole fundamental constituent; spacetime, fields, and matter are emergent. From this replacement, the phase-saturated black hole emerges as the natural consequence of accumulated phase inertia beyond the critical threshold — a new phase of the phase substrate, stable, structured, and entirely lawful.

The resolution of black hole paradoxes is not merely a technical achievement within this framework. It is a confirmation — the strongest confirmation yet obtained — of the ontological claim of Phase Theory: that the universe is, at its most fundamental level, a phase substrate governed by admissibility. Spacetime is a story we tell about phase correlations in the low-curvature regime. Black holes are the story breaking down. Phase saturation is what is actually there.

The work ahead is clear. Paper 14 will extend phase saturation to the cosmological setting, reformulating the Big Bang as a global phase desaturation transition and replacing the initial singularity with the cosmological phase core. Paper 15 will apply the near-saturation formalism to neutron stars and compact objects, deriving the neutron star equation of state from phase inertia accumulation dynamics. Papers 16 and beyond will address phase-theoretic treatments of quantum measurement, the measurement problem, and the emergence of the classical world from phase topology.

The program is unified. The foundations are established. The observational predictions are enumerated and, with advancing gravitational wave astronomy, very-long-baseline interferometry, and analogue gravity experiments, they are becoming testable. Phase Theory stands ready for the verdict of observation.

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Sections 5, 12, and 18 draw directly on technical results from Paper 12 (Quantum Gravity Foundations). The thermodynamic framework of Sections 8 and 10 rests entirely on Paper 8 (Phase Thermodynamics). The author thanks the broader community of researchers in quantum gravity, black hole physics, and quantum information whose decades of work — including the formulations that Phase Theory here transcends — provided the precise targets against which a new ontological framework must be measured.

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